lyra-cpp — Session Log

Running EOD log. Newest entry on top. Short rough-outline format.

2026-06-29 (EOD) — Post-v0.7.0: docs + VAC latency posture (VarAC turnaround)

All on `main`, unreleased (after v0.7.0). A doc pass + one operator feature.

Docs (`e199993`)

• Recorded the v0.7.0 **FM transmit refinements** as a TX-relevant entry in

`THETIS_DIRECT_PORT_PLAN.md`'s SHIPPED-WORK LOG (FM is a TX modulator concern): fix #2 AF brick-wall `b72bc50`, FM-aware bandwidth `4f32ca1`, pre-emphasis Off/Comm `a826c8c`, mic-rack auto-bypass `12c8991`, Tuning-panel pre-emph chip + Dev→RX-BW `67abf05`. RESUME POINTER + trailer refreshed to v0.7.0 (PureSignal + RX2/Split = remaining major TX-port arcs). Regen plan PDF/DOCX.

• New `docs/REMAINING_WORK.md` (+ PDF): standalone what's-left checklist — much

smaller now (two big arcs left, RX2/Split + PureSignal, plus TX/CW follow-ons). Cleaned two stale tasks (CW macros #176, Spots #182 — shipped).

VAC latency posture — `dc9c4c2` (#158 follow-up, operator-bench-confirmed "works better for VarAC")

• Operator wanted to squeeze VAC TX↔RX turnaround for VarAC/ARQ (runs 25 ms

rings in Thetis). The IVAC engine already ported the full latency surface (`SetIVACInLatency`/`OutLatency`/`PAInLatency`/`PAOutLatency` + `vac_size`, driven by hardcoded `vac1LatencyMs_=120`/`vac1VacSize_=2048`); this exposed it.

• **Settings** → **Audio** → **VAC1** gains: **Buffer size** dropdown (128..8192, each

ms-labelled at 48 kHz), **Latency** spinbox (5..500 ms — rmatchV ring depth; one knob feeds all four reference latency fields), + a live **Monitor** line (ring fill % + overflow/underflow per direction, from `getIVACdiags` on a 500 ms QTimer — the reference VAC1 Monitor). Drop latency until counters twitch, back off.

• `setVac1LatencyMs`/`setVac1VacSize` persist (`vac1/latencyMs`, `vac1/vacSize`)

+ reopen VAC when live via the proven `rebuildVac1` path; `vac1Diags()` = guarded diag read.

• **Carried per-profile (Profile schema v4→v5, tolerant-load):** a tight VarAC

profile keeps low latency while SSB stays at 2048/120 — flip profiles, flip the whole VAC posture. USER_GUIDE "Latency (for fast ARQ modes like VarAC)" note added.

• GOTCHA: reference `ivac.c:795` type comment is REVERSED vs its code; the

TO/FROM VAC diag labels follow the code (`rmatchOUT`=TO type0 / `rmatchIN`=FROM type1, Ivac.cpp:797). Flag if they read backwards at bench.

• Operator decision: **ACCUMULATE** — stays on `main` unreleased; testers stay

on v0.7.0 until the next release bundles it.

NEXT SESSION — operator-flagged (2026-06-29 EOD)

1. **Tuner panel delete** → **trashcan icon.** Cosmetic: the per-row delete control

on the Tuner panel (currently the − button, with the ✓/✗ confirm) should be a trashcan symbol. `src/qml/TunerPanel.qml`.

2. **BUG — layout save/recall re-adds the Solar panel.** Saving a named panel

layout WITHOUT the Solar (HamQSL) panel open, then recalling it, ADDS the Solar panel even though it wasn't present at save time. **Only the Solar panel shows this so far** — other panels round-trip their hidden state fine. Smells like the Solar panel has a default-visible/auto-show that the layout-restore path doesn't suppress (cf. the Tuner base-dock "shows on summon" pattern from this session — a dock that buildDocks creates visible/auto and the saved hidden state doesn't override on restore). Investigate the named-layout save/restore (#184 / v0.4.8) dock-visibility serialization for the Solar/HamQSL panel specifically. See [[reference-hamqsl-solar]].

3. **■ SAFETY BUG (task #190) — USB-BCD adapter intermittently undetected +

amp changed bands during RX.** FTDI C232HM (FT232H, COM1 in Windows) often not seen by Lyra's BCD dropdown; amp "beeped + changed bands in RX" (NOT RFI — heavily choked cable). Code-read root causes in `src/usb_bcd.cpp`: (a) enumeration is D2XX-only and the adapter is in VCP mode (owns the interface) → intermittent D2XX enumerate (the unplug/replug symptom); (b) `devices()` (~103-114) filters empty serials, but FTDI returns an empty serial for any OPEN device and Lyra's ctor opens+holds the device when BCD is enabled → its own dropdown shows "(none)" (matches the screenshot). Fix on the operator's bench (verify-don't-guess; spurious-BCD glitch is a separate lower-confidence item — get a repro + logging first). Interim safety: untick "Enable USB-BCD amp band output". Full detail in the memory note + task #190.

2026-06-29 — RELEASED v0.7.0 "Tuner memory" (Tuner panel + FM polish + tooltips toggle + header tidy)

Cut **v0.7.0** off `main` (bump `CMakeLists.txt` + `installer/lyra.iss` 0.6.0 → 0.7.0; README Features → v0.7.0 + Tuner bullet; release note `docs/releases/v0.7.0.md`; USER_GUIDE "Tuner (manual ATU memory)" section + header-chip + TOC; regen PDF/DOCX). Installer `dist/Lyra-Setup-0.7.0.exe`.

Headline: the new **Tuner** manual-ATU memory subsystem (operator-requested 2026-06-29, benched this session).

• `src/tunermemory.{h,cpp}` + `src/qml/TunerPanel.qml` + Settings → Tuner tab.

3 antennas × freq-keyed Input/Output/Inductor, live colour-coded SWR, nearest-point **bracket** view (1 each side; exact + 1 each side when on a point — `windowEach=1`), Store→Save, single-row inline edit, ✓/✗ delete confirm. Match by carrier freq (USB/LSB-agnostic). Pure UI + QSettings, RF-safe.

• Tuner polish this session: SWR pill enlarged; flat (background-`Item{}`)

header buttons; `matchDeltaHz` sign flipped to `point − dial` (so **+** = saved point above the dial — read the natural way).

• Header chip strip regrouped: **TX DSP** · **RX DSP** · **Options**

(CTUN · Tuner · CW · CW Dec · WF-ID). CW/CW Dec/Tuner moved into Options; RX DSP slid left. `mainwindow.cpp buildToolbar`.

• Also folded into v0.7.0 (committed since v0.6.0): FM pre-emphasis Off/Comm

selector + FM-aware bandwidth + Deviation→RX-BW auto-size + native mic rack auto-bypass on FM; global **Show tooltips** toggle (Settings → Visuals) + Profile-picker polish.

• The earlier "float like TX DSP" experiment for the Tuner dock was reverted

(`11dd6d2`) after it caused intermittent launches; re-approached as just the `isChipSummonedPanel` predicate entry (`297a753`) which fixed the cyan dock-drop targets without the launch issue.

2026-06-29 — FM TX "beat Thetis" arc (pre-emphasis + clean chain + Tuning-panel UX)

All on `main`, unreleased (on top of v0.6.0). The FM transmit arc, brought to a complete stopping point. WDSP study-first: read `fmmod.c` + `emph.c` in full before touching shipped behaviour.

FM Fix #2 — clean audio band-limit (`b72bc50`, earlier; logged for completeness)

• The FM primary bandpass `bp0` runs on the AUDIO before `xfmmod`, so it's

now a symmetric **±3 kHz brick-wall AF low-pass** (was inheriting ±(dev+3k) ≈ ±8 k, over-feeding the modulator). The occupied-RF (Carson) clamp stays owned by the modulator's own internal bandpass, pinned via new `SetTXAFMAFFreqs(txch, 300, 3000)` on FM entry. Supersedes Fix #1's bp0 value (v0.6.0).

Mode-aware FM bandwidth (`4f32ca1`, earlier) — `ModeFilterPanel.qml`

• FM TX-BW selector + RX↔TX ■ lock disabled in FM (FM width is derived from

deviation). TX-BW box is a read-only **" k (auto)"** Carson readout `2·(dev+3k)`, live-tracking Dev. FM RX presets = FM channel widths

8/10/12/16 k (was the SSB list).

#3a — pre-emphasis selector Off / Comm (`a826c8c`)

• WDSP-native via `SetTXAFMEmphPosition` (bound in wdspcalls X-macro):

Comm = position 1 = the native 6 dB/oct (300–3000) comms curve runs;

Off = position 2 = both call sites pass through = a TRUE bypass (FM force-runs the emph block, so the chain position is the on/off — a real Off, which rigs/Thetis don't give you).

• `HL2Stream` `fmEmphasisMode` Q_PROPERTY (0=Off, 1=Comm, default Comm) +

QSettings `tx/fmEmphasisMode` + `TxControl::setFmEmphasis` callback; applied on FM entry in `pushTxFilter` + live on operator change.

• UI: **Settings** → **TX** → **FM** combo (beside Deviation / CTCSS).

#3b-3 — FM auto-bypasses the native mic rack (`12c8991`)

• In FM the whole TX rack (EQ + Combinator + Plate + speech enhancements)

auto-bypasses, same forced-auto as the digital modes — a multiband compressor/reverb into FM's pre-emphasis = mush. One-line predicate change in `main.cpp` (`txModeBypassesRack` now fires for "FM" alongside DIGU/DIGL → `SetTxRackBypass`).

Tuning-panel UX (`67abf05`) — `TuningPanel.qml`

• **Emph** quick chip (Comm / Off) added to the FM front row beside Dev/RPT,

two-way-synced with Settings via the shared `Stream.fmEmphasisMode` prop.

• **Deviation now auto-sizes RX bandwidth** to the smallest FM channel

preset ≥ Carson `2·(dev+3k)` — ±2.5 k → 12 k, ±5 k → 16 k. TX BW was already deviation-derived; RX now follows (Thetis-style). Still overridable in the Filters panel.

#3b 50/75 µs + 300 Hz HPF — DROPPED / DEFERRED (operator call)

• Read `emph.c`: WDSP FM pre-emph is a 6 dB/oct ramp between f_low/f_high.

50/75 µs are broadcast-FM constants (15 kHz audio, ±75 kHz dev), meaningless over our 3 kHz comms band (75 µs corner ≈2122 Hz = a weak top-end lift; 50 µs corner ≈3183 Hz is *above* the 3 kHz cutoff → no pre-emph at all). "Comm" already IS the correct amateur curve → 50/75 µs

dropped (dead UI options). **300 Hz low-cut deferred** — needs a new native HPF stage (can't ride bp0 = Hilbert trap; rack is bypassed in FM), mostly redundant with Comm's bass rolloff; build only if FM-data-flat bass-cut is later wanted.

Docs

• USER_GUIDE: FM section rewritten (deviation / pre-emphasis / CTCSS +

clean-chain note + Tuning-panel mirror + Dev→RX-BW); TX-BW + Lock bullets note the FM auto-width behaviour; Tuning-panel FM front-row bullets updated.

UI cosmetics — global tooltips toggle + Profile picker (`ff0cdf9`)

• **Global "Show tooltips" toggle** (Settings → Visuals, default on,

persisted `ui/tooltips_enabled`). New `Prefs.tooltipsEnabled`; all 69 QML `ToolTip.visible` bindings across 9 panels AND it in (swept), and an app-wide `QEvent::ToolTip` event filter (`TooltipGate` in mainwindow.cpp) swallows QWidget (Settings dialog / menu) tooltips too → one genuinely global switch. USER_GUIDE Visuals → Tooltips section added.

• **Profile picker (ProfilePanel):** the recall tooltip no longer pops over

the open dropdown (gated `!popup.visible`); combo width 300 → 170.

2026-06-28 — v0.6.0 RELEASED: per-band TX power model + auto-calibrated watts cap + FM fix #1

Released `v0.6.0` (`5b41fc3`; installer `dist/Lyra-Setup-0.6.0.exe`, 68.2 MB, GitHub release published; main FF'd). Bundled the whole TX power-model arc + the first FM fix that had been pending on main.

Per-band TX power model (Settings → PA Gain tab)

• ChannelMaster `txgain.c` ported → `wire/TxGain.*`; operator drive% drives a

digital fixed gain (Thetis RadioVolume) on top of the AD9866 byte, so **1 % drive is a true sub-watt** (was pinned at the ~0.78 W nibble-0 dead zone).

• Per-band **PA Gain table** (`RadioVolume = min(drive·gbb[band]/100/93.75,1)`)

so the dial can be tuned to read true watts per band.

• **Watts Max cap** (amp protection) + **auto-calibrate**: key TUN per band

into a dummy and Lyra walks power up to the cap and locks that band exactly (✓), SS-amp-safe (approach-from-below). Coloured Cap-tuned marks (green ✓ tuned / red — not tuned). The old "Max TX drive %" (#170) control retired.

FM fix #1 (`ae6eb19`) — FM Carson bandpass clamp (value later superseded by Fix #2)

• FM output bandpass sized from the FM signal's own occupied channel instead

of inheriting the wide SSB/ESSB TX width → no FM splatter at wide TX BW.

Docs

• USER_GUIDE "Settings → PA Gain" section; `docs/releases/v0.6.0.md`; README

bumped to v0.6.0.

2026-06-27 — v0.5.0 RELEASED: DX-cluster spots subsystem (#182) + CW RX-decoder overhaul (#187) + #159 DSP filter type

Released `v0.5.0` (`63b0799`; CW `7ebc599`) — two commits on `main`, installer `dist/Lyra-Setup-0.5.0.exe` (68.2 MB) published as a GitHub release. (v0.4.11 — the Kenwood CAT-over-COM/TCP + serial-PTT arc — shipped between v0.4.10 and this; it has its own release notes.)

DX-cluster spots subsystem #182 — the headline (`spotstore.*`, `spothole_feeder.*`, `dxcluster_feeder.*`, `dxcc.*`, `PanadapterPanel.qml`, `settingsdialog.*`)

• Source-agnostic **spot bus** (`SpotStore`): call-keyed dedup, age-out,

own-call highlight + toast, `source/continent/firstSeen` fields.

• **3 feeders**, each its own on/off (Settings → Network → DX-cluster spots):

  • **TCI** inbound (logger / SDRLogger+) — `addSpot(...,source="tci")`.

  • **SpotHole REST** (`SpotHoleFeeder`) — `spothole.app/api/v1/spots`,

standalone (internet only), default-OFF poll. Parse is **snake_case** (`dx_call` / `freq` in **Hz** / `de_call` / `dx_continent` / skip `qrt`) — first cut used hamlog's processed field names and got zero spots; fixed by probing the raw API.

  • **DX-cluster telnet** (`DxClusterFeeder`) — host/port/login-call

(defaults MYCALL, overridable) + login-commands, auto-login + auto-reconnect 20 s. NOTE: VE7CC-type nodes default to **RBN CW skimmer only**; `set/ft8;set/ft4` login cmds open up FT8/FT4 (confirmed by reading the node banner — not a bug).

• **Filters** (display-only; the bus keeps everything): `modeShown`

(family-aware CW/SSB/DIGI/AM/FM), `regionShown` + `DxccLookup::continentOf` ISO-2→continent table (continent codes NA/EU/AS/SA/AF/OC always mean continents, else ISO-2 country; "US,EU" = US + all Europe).

• **Clutter caps**: per-freq bucket cap (`freqHz/bucketHz` bins, keep N

newest, "+K more" badge — tames FT8 piles) + global display cap.

• **Colour modes**: source / single / by-mode / by-region / by-country +

optional legend (`legendEntries()` Q_INVOKABLE). **New-spot colour**: a fresh spot wears a solid colour for N s (driven by `firstSeen`, not `added`, so a re-poll doesn't reset it) then settles. Two bugs fixed here: spots flipping green↔gold every poll (separated `firstSeen` from `added`), and the original blink flashing EVERYTHING on startup (reverted blink → solid new-spot colour;

renamed all "Flash" UI + docs → "new spot colour" so folks don't expect a blink).

• **Console import**: click a spot → tune to it + `CwMacros.hisCall` grabs

the call into the CW-console contact row (#176/#181).

• Band display filter DEFERRED (redundant on the panadapter; lands with a

future spot-LIST view).

CW RX-decoder overhaul #187 (`CwDecoderPanel.qml`)

• Decode-window **scroll fix**: ScrollView + `onContentHeightChanged →

Qt.callLater(scrollToBottom)` so rows stop clipping at the 3rd line before a resize. (Took two passes — the first "fixed" it only until the text reflowed.)

• Narrow detect-filter BW chips tuned to `[80,100,120,150]`; Auto level

with live readout; Advanced (Seek / DSP filter / AFC range). Operator copy-confirmed on real Field Day CW (S4–S9, 10–25 wpm).

Also shipped

• **CW console scroll** (`CwConsolePanel.qml`): wrapped `body` in a

ScrollView so the macro tags + bottom option rows stay reachable when the console is sized small.

• **#159 DSP filter type** (Linear Phase / Low Latency per mode family,

opt-in) — was the prior pending item; rode out with v0.5.0.

Docs / release

• USER_GUIDE: new "DX-cluster spots" section + reworked "Reading CW — the

RX decoder" section; #159 Linear Phase / Low Latency already present.

• `docs/releases/v0.5.0.md`; version single-sourced via CMake

`project(VERSION 0.5.0)` → installer `.iss` + README.

• Installer `Excludes` now drops `test_*.exe` (release hygiene).

2026-06-25 — v0.4.10 RELEASED: CTUN drag-the-marker (4th look, FIXED) + CW decoder A-series + CW-decoder docs

Working on `main`. Two releases since v0.4.8: **v0.4.9** (fast patch — floating tool panels are float-only, Lock no longer disables the CW console/decoder; `162906f`) and today's **v0.4.10** (`ebb3c67`).

CTUN #174 — 4th look, the pass that actually fixed it (`PanadapterPanel.qml`)

• Operator (Thetis Help open as ground truth) said CTUN still wasn't Thetis

after the v0.4.4/v0.4.5 "parity" rebuilds: you should drag the FREQ MARKER onto signals with the display LOCKED; a click INSIDE the filter must NOT jump the 0-beat.

• Read the real Thetis mouse handlers FIRST (console.cs 50230-50433 press +

51887-51960 drag). The DSP core (locked-LO + WDSP NCO shift) was already right; the whole gap was the MOUSE + MARKER layer.

• ROOT CAUSE the 3 prior passes all missed: Lyra's carrier marker is

hard-pinned to the locked display centre (`markerOffsetHz` = CW pitch only), so under CTUN there was NO marker tracking the dial — nothing to drag. First bench report: "marker does not move."

• Fix (CTUN-gated, non-CTUN byte-identical): new `ctunDialOffHz` slides the

carrier marker + filter passband to (rx1FreqHz – centre); body-drag = relative 1:1 marker-grab (`setRx1FreqHz`); top-15px strip = LO pan (`setCtuneCenterHz`); inside-passband click suppresses the retune; the edge-drag + inPassband math subtract the dial offset.

• Operator-bench-CONFIRMED "Do as Thetis does state." #174 DONE.

CW decoder — rest of the safe Tier-A A-series (A4/A6/A7/A8) (`CwDecoder.cpp/.h`)

• A-slice A1/A2/A3/A5 was committed last session (`eb4c89d`). v0.4.10 adds:

• A4 sub-bin AFC: 3-pt parabolic interp on the winning Goertzel bin

(refines bestHz only).

• A6 hysteretic SNR gate: open at squelch×floor, hold to 0.8× that

(≈1.2× floor); inert on clean signal.

• A7 unify the two word-gap emitters: guard the processEdge Word emit with

`lastEmitted_ != ' '` so a word gap can't double-emit.

• A8 add _ / `$` punctuation; prosigns deliberately omitted (alias +/( ).

• All additive — correct decodes unchanged, ~35 tuned constants untouched.

Unit test ALL PASS (VVV TEST CQ / CQ PARIS / CQ TEST; AFC lock 749).

CW decoder docs (`USER_GUIDE.md`)

• Operator couldn't recall what the decoder NB / DSP-filter did → realised

none of the decoder was documented (+ a stale "decoder coming in a later release" line). Added "Reading CW — the RX decoder" section + TOC entry: panel, confidence dimming, decoded-call/name grab → His Call / Name (feeds {CALL}/{NAME}), Match-TX-to-RX-WPM, and every detector knob. Console His-call cross-linked to the decoder grab.

B1 anchor (NEXT CW work — NOT started)

• Benched a live **W1AW** run; it announces its own speed ("END OF 30WPM

TEXT", "NOW 25 WPM") = labeled truth. Sent 30→25 wpm, decoder read **35** → over-reads ~15-20%, dit estimate runs short. Copy still readable; residual errors (WILL→WIRL, ONCE→JNCE) are valid-letter swaps A1 can't touch by design. W1AW = the labeled bench to tune B1 (independent dah-length + warm-up/Gaussian boundary fix).

Release mechanics + standing

• 0.4.9 → 0.4.10 (CMakeLists / lyra.iss / README); notes

`docs/releases/v0.4.10.md`. Commit `ebb3c67`, tag `v0.4.10`, pushed main + tag, GitHub release + `dist/Lyra-Setup-0.4.10.exe` published.

• Daily backup task ("Lyra-cpp Daily Backup", 22:30 → `_backups/` + mirror

`Z:\LyraBU\`) confirmed ACTIVE + healthy (last run 6/24 OK, 0x0).

• Open: CW B1 (with the W1AW bench). CTUN done. Bigger arcs still open —

RX2 #96-101, CW macro chips #176, grab-spot-as-call #182, PureSignal.

2026-06-24 — v0.4.8 RELEASED: panel docking overhaul + CW transmit metering (+ wiki, CW-decoder review)

Working on `main` (last tag `v0.4.7` = `ff6c9c9`). Tester feedback (Randy W7CPA / Jim / a 3rd) said the main panels "all float, nothing docks, drop-against-an-edge does nothing," and resizing was a fight. Big UI arc + a CW metering arc, shipped as **v0.4.8**.

Panel docking overhaul (#184) — `dockdragcontroller.{h,cpp}` (NEW) + `mainwindow.{h,cpp}`

• Root cause: the custom dock title bar (`makeDockTitleBar`) replaces

QDockWidget's built-in drag, killing Qt's drag->drop-indicator->snap-to-edge.

• New `DockDragController` (public QMainWindow API only): press->threshold->

drag->drop FSM on each title bar, translucent **cyan drop-zone overlay** — snap-to-edge (`addDockWidget`), neighbor split L/R/T/B thirds (`splitDockWidget`, "before"=double-split), center tabify (`tabifyDockWidget`), off-region float (`setFloating`). Save-on-commit.

• Resize: dock separators widened 6px + cyan-on-hover; tab bar styled.

Compact custom move cursor (arrow while locked). One-time status-bar hint.

• **Lock = move + resize**: disables move/float/close via features AND blocks

the dock-separator resize press in `event()` (split-cursor detect). The earlier `setFixedSize`-on-lock froze panels and collapsed the layout on a locked restart — REMOVED; event-interception locks resize cleanly and always restores. (Two intermediate restart-collapse symptoms chased + fixed.)

• View -> Layouts: **4 named slots + Lyra default** (recall no longer

force-shows the chip-summoned TX/RX/CW rack panels).

Factory default fixed — `default_layout.h` *+* `applyDefaultLayout`

• Prior embedded default was base64 of the raw registry REG_BINARY -> carried

the QSettings `@ByteArray(...)` UTF-16 wrapper -> `restoreState()` rejected it (the "screwed up default"). Re-embedded from operator's **N8SDR** slot, de-wrapped to the real saveState bytes, + `defaultPanadapterSplit()`.

CW transmit metering (#105) — `hl2_stream.{h,cpp}`*,* `metermodel.{h,cpp}`*,* `main.cpp`*, QML*

• Bug: QSK CW keys the PA at the gateware and deliberately leaves wire MOX

low, so the meter (gated on `moxActive`) never flipped to TX during CW (external watt-meter showed power). Other modes flip because they assert MOX.

• `cwKeyingActive` (message-level, from the CwKeyer `onState` hook, marshalled

to the stream thread) drives the meter RX<->TX swap + the Ladder source set. Reads `fwdPowerW()` (already valid during gateware CW). Does NOT assert wire MOX — QSK keying unchanged.

• `txDisplayActive` (= `moxActive || cwKeyingActive`) drives the **red-on-air**

QML (VFO border + RX/TX tag in TuningPanel, split/XIT TX markers in PanadapterPanel).

• **Paddle/key** detection: `onHwPttPoll` (~50 ms) latches `cwKeyingActive`

from **forward power** (>=0.5 W, ~500 ms hang), gated to CW mode, skipped while CWX drives it — this HL2+ reads non-zero `ptt_in` at rest (§10 Q#1), so forward power is the clean signal.

• The panadapter spectrum **stays RX** during CW (gateware CW never feeds the

WDSP TX analyzer -> swapping shows an empty pan; RX shows the keyed carrier). Operator-confirmed regression + fix.

Release / docs / wiki

• Version 0.4.7->0.4.8 (CMake `project(VERSION)` + `installer/lyra.iss`).

`docs/releases/v0.4.8.md`; USER_GUIDE "Getting around the window" rewritten + Meter CW note; README Features->v0.4.8. ISCC -> `dist/Lyra-Setup-0.4.8.exe` (69 MB). Tag `v0.4.8`, pushed main+tag, GitHub Release published w/ installer.

• **GitHub Wiki initialized + built**: Home, User Guide (mirror of

USER_GUIDE.md), Installing-and-first-connection, FAQ-and-Troubleshooting, _Sidebar. Sources version-controlled in `docs/wiki/` (User Guide generated from USER_GUIDE.md at publish). NB: the first wiki page must be created once in the web UI (no API, no push-init) — operator did it; rest pushed via `…wiki.git`.

CW decoder accuracy review (analysis only — NOTHING changed)

• Recovered the 4-reviewer + code-cross-checked review (was transcript-only)

and committed it: `docs/CW_DECODER_REVIEW.md` (Tiers A–D). Target = `src/dsp/CwDecoder.cpp/.h` (field-tuned constants).

• Operator chose the safe **A-slice = A1+A2+A3+A5** to do next (nearest-code

`?` rescue / AFC center-bias / per-char confidence / crash-proof peak-snap — none touch the tuned constants; rule: never corrupt a correct decode). Then **B1** (independent dah-length + the √3-vs-1.565 boundary discontinuity). NOT STARTED. See [[project-lyra-cpp-cw-decoder-accuracy]].

2026-06-23 — v0.4.6 RELEASED: connectivity + multi-radio + front-panel polish

Working directly on `main` (last release v0.4.5 `4a106d5`). Connectivity/ polish release — no wire/TX/audio-path changes; all UI / discovery / installer.

Installer (lyra.iss)

• Auto Windows Firewall **inbound** rules for `lyra.exe`: UDP (HL2 discovery

reply + EP6 RX stream) + TCP (TCI server, future networked clients). Named "Lyra SDR", delete-then-add (idempotent), removed on uninstall. **Fixes the "needs admin to connect" report** — the unsolicited inbound discovery reply was being dropped for a non-elevated launch. Matches Thetis's WiX FirewallException approach.

• Opt-in `NetworkThrottlingIndex=0xffffffff` task (default UNticked) — the

classic multimedia network-throttle "gaming" tweak; `[Registry]`, left in place on uninstall. Mirrors Thetis `NetworkThrottle.cs`.

Discovery (hl2_discovery.{h,cpp})

• **Dual broadcast**: `sendBroadcastFromAllSockets` now sends to both

`255.255.255.255` AND each NIC's subnet-directed broadcast (`entry.broadcast()`).

• `probe(ip)` — directed unicast discovery for Add-by-IP (own socket +

1 s deadline, reuses parseReply→radioFound). Independent of the sweep.

Settings → Hardware radio panel (settingsdialog.cpp)

• **Add by IP** fires `probe()` → the placeholder upgrades in place to the real

board/gw/rx (addRadio changed skip-on-dup → update-in-place).

• **Double-click a row** → **Open**; **connected radio shown green + bold**

(item roles, survives re-render); one-line helper.

Front panel (mainwindow.{h,cpp})

• **Start green / Stop red** (backing QToolButton), **Connected green /

Disconnected red / Connecting-Scanning amber** (connStatus_).

• Header chip strip: new **Options:** separator+label so **CTUN** / **WF-ID**

no longer read as RX-DSP functions.

Release

• Version bump CMakeLists `0.4.5`→`0.4.6` + lyra.iss AppVersion. Release note

`docs/releases/v0.4.6.md`; USER_GUIDE header + Radio sections; README Features (v0.4.6). Tag `v0.4.6`, installer rebuilt, pushed to origin/main + GitHub release.

2026-06-22 — v0.4.5 RELEASED: CTUNE rebuild + Waterfall ID + CW macros

Branch `feature/ctune-rebuild-wfid`; `main` was at v0.4.4. This is the v0.4.5 release.

#174 CTUNE rebuilt (operator: v0.4.4 "built wrong")

• 4-agent Thetis-vs-Lyra audit. Centre/shift/edge decision relocated from QML

into the C++ RX-NCO chokepoint (`HL2Stream::pushEffectiveRxFreq`) so EVERY tune source behaves (dial / click / keypad / band+memory / TCI spot): smooth- scroll at the 5% display margin, far-jump re-centre, zoom force-centre, IQ- Nyquist clamp. `RXANBPSetShiftFrequency` added so notches/NB track the shift.

• Commit `29acc6d` (design docs `0e0f2f2`).

#175 Waterfall callsign ID

• TX courtesy ID: the call rendered as a readable image in the SSB passband.

Header arm chip (USB/LSB only), Settings → TX config (level ≤0.065 / repeat / over-drive warning), crash-safe transient flat-TX orchestrator (TCI source + whole-rack bypass + DIGU/DIGL, restore on un-key).

• **LSB mirror bug fixed** this session: DIGL flips audio↔RF, so the call came

out mirrored in LSB (USB was fine). Renderer now pre-mirrors the frequency layout for LSB → reads upright on both sidebands. Commit `79d8d43`.

#176 CW macros — named click / F-key memories

• Redesigned floating CW console into a contest-grade macro bank: named click +

F-key (F1–F12, global) memories, a "My macros" group the operator builds out, live "sending" state, Repeat, a contact row (His call / Name / RST / #).

• Tokens: contact `{CALL}/{NAME}/{RST}/{#}` + `{MYCALL}`, plus operator personal

tokens (`{ME}/{PWR}/{QTH}/{RIG}/...`), with a click-to-insert palette.

• New `CwMacroModel` persists macros / tokens / serial to QSettings. F-keys via

`MainWindow::keyPressEvent` (QML Shortcuts don't fire in the QQuickWidget docks). Commit `9f0d054`.

Settings dialog

• Per-tab scroll-area + larger default size (long tabs were clipping).

Release

• Version bumped (`CMakeLists.txt` + `installer/lyra.iss` → 0.4.5), release note

`docs/releases/v0.4.5.md`, User Guide CW section updated, annotated tag `v0.4.5`, installer `dist/Lyra-Setup-0.4.5.exe`, pushed to GitHub; `main` fast-forwarded to the release.

2026-06-21 — v0.4.3 RELEASED: 60 m band plan + tune-drive modes + BCD polish

Operator quality-of-life batch + release. All on `main` (project commits releases directly to `main`, tags + GitHub release + installer). Released mid-afternoon, then held for the day.

60 m band plan (`bandplan.{h,cpp}`, `PanadapterPanel.qml`)

• **Clickable 60 m channels**: US CH1–CH5 now render as always-visible

labelled bars (a touch larger), each a MouseArea that QSYs (`Prefs.mode= "USB"` + `Stream.setRx1FreqHz(qsy)`) to the channel's USB dial — same one-click idiom as the FT8/FT4/PSK watering-hole markers. `segments()` flags `channel`/`qsy` per sub-band; delegate gained `chan` prop + conditional label size/visibility + tooltip.

• **Region/country-correct 60 m**: precise US channel dial freqs

(5330.5/5346.5/5357.0/5371.5/5403.5 kHz); WRC-15 contiguous band (5351.5–5366.5) added to `r1Bands()` (was missing). New **Country** override layer: `regionBands(region)` + `countryBands(country)` + `bandsFor(region, country)` merge (replaces same-named band in place). `BandPlan::country()` + `bandPlanCountryChanged→regionChanged`; all 3 `bandsFor` call sites updated. UK = its 11 RSGB/Ofcom segments; Canada = WRC-15. `Prefs.bandPlanCountry` (key `band_plan/country`, default "AUTO", validated vs {AUTO,UK,CA}). Settings → Hardware Country combo added.

• Verify-don't-guess applied: UK 11-segment list WebFetch-confirmed

(rsgb.org); AskUserQuestion before writing international freqs.

Tune-drive selector — #95 (`prefs`, `main.cpp`, `Profile`, settings, TxPanel)

• Replaced bool `useTuneDrive` with enum `tuneDriveMode`

{Slider=0/Tune=1/Fixed=2} + `fixedTuneDrivePct` (default 25), one-time QSettings migration (legacy bool → 0/1). main.cpp tune orchestrator rewritten to a `tunePctFor()` lambda + mode-aware arm/release (fixes a latent mode-switch-mid-tune restore edge). Profile capture/apply + TxPanel `visible: Prefs.tuneDriveMode === 1` updated.

• **Max-drive cap is honored**: wire chokepoint `setTxDriveLevel` (`std::min(maxDriveRaw(), …)`) already covers TUN; added cosmetic defense-in-depth — Tune/Fixed spinners `setRange(0, maxDrivePct())` + live follow on `maxDrivePctChanged`. So TUN can't exceed the operator's Max-TX-drive ceiling.

11 m → 10 m BCD option (`usb_bcd.{h,cpp}`, settingsdialog)

• New `elevenAsTen` (key `hw/bcd11as10`, default ON), mirroring the

existing `sixtyAsForty`. Substitution lives in `reapply()`: an 11 m/CB freq (`cbBandIndexForFreq`>=0, amateur table returns -1) remaps to the 10 m amateur index → emits **BCD 9**. Settings → Hardware checkbox.

• Plus an amber regulatory advisory under the BCD options: "Operate only

within the maximum power and band limits permitted by your country / region's regulations." (Operator-requested — states it plainly at the point of use re: 11 m abuse.) USER_GUIDE BCD section mirrors it.

Docs + release

• USER_GUIDE: Band-plan (Region+Country+60 m channels), Settings → TX

(Tune-drive subsection + Max-drive cap note), USB-BCD (11 m option + advisory). New `docs/releases/v0.4.3.md` (= GitHub release body).

• Version 0.4.2 → 0.4.3 (`CMakeLists.txt` + `installer/lyra.iss`); full

rebuild (LYRA_VERSION embedded, USER_GUIDE resource recompiled) + ISCC → `dist/Lyra-Setup-0.4.3.exe` (68 MB).

• `06921f2` Release v0.4.3 (16 files, +598/–158) + the already-on-main

help-doc commit `4e62bb9` (log-capture section) rode out together. Tag `v0.4.3`, pushed `main` (`ddd6677..06921f2`) + tag, GitHub release published with installer attached + verified.

Standing rules in effect

• NO PATCHING (tune-mode done as a proper enum + migration, not a shim).

• Commit/push only when asked — operator explicitly requested this release.

• Verify-don't-guess for regulatory data (60 m freqs, UK segments).

2026-06-19 — v0.4.1 RELEASED: TX protection (SWR cut + drive cap) + PHROT

Short session: shipped the TX-protection arc, a PHROT parity feature with a digital auto-gate, doc updates, then cut + pushed **v0.4.1** (main FF'd, tag + GitHub release + installer). All on `main`.

TX protection — #169 SWR + #170 drive cap (SHIPPED + operator-bench-confirmed)

• **#169 SWR Cut** (`6c3e8e7`): a 50 ms QTimer (`swrEvalTimer_`) armed on the

MOX edge in `HL2Stream` decodes `prn->tx[0].fwd/rev_power` → ρ=√(rev/fwd), SWR=(1+ρ)/(1–ρ) (calibration-free). Trips through the single `requestMox(false)` unkey funnel above a user threshold (default 5:1). Four false-trigger guards: blank window, fwd/rev power floors (NaN-safe), dwell. Latch + reason + **PROT lamp** on TxPanel (gray/green/red, elided so it can't overflow the ATT lamp). Settings → TX group (enable + limit + Cut/Fold + fold-floor + during-tune).

• **#169 Phase 1b Fold**: optional action — steps drive ×0.5 via

`applyDriveLevelNoPersist` (no QSettings write, fold restored next keydown) and escalates to a hard trip at the floor.

• **#170a drive cap**: `maxDrivePct` (1..100%) clamps every drive write at the

one chokepoint `setTxDriveLevel` (`std::min(maxDriveRaw(), …)`) — covers the operator slider, TUN, BandMemory restore, TCI DRIVE alike, + re-clamps live on lowering. Operator bench-confirmed ("ran slider max down, couldn't pull more"). **#170b over-power trip DEFERRED** by operator ("fine as it is").

• Design doc `docs/architecture/tx_protection_design.md` stamped COMPLETE.

PHROT — #109 phase rotator toggle + digital auto-off (SHIPPED)

• WDSP TXA `SetTXAPHROTRun` exposed via the `wire/wdspcalls` X-macro seam +

a `TxControl.setPhrotRun` callback (`d99794d`). `phrotEnabled` Q_PROPERTY (Settings → TX checkbox, default ON = WDSP/reference posture). Parity with the reference Setup PHROT control — it is fully exposed there, not a hidden knob (operator asked; verified).

• **Auto-off in digital** (`165e176`): one chokepoint `applyPhrotRun()`

computes `run = phrotEnabled_ && !(txMode==DIGU|DIGL)` (WDSP mode 7/9) and pushes via the callback — called from the operator setter, **every** `setTxMode` **edge**, and `registerTxControl` (channel open). The checkbox is now the operator's *voice-mode* intent; DIGU/DIGL switch it off on the wire and back on in voice modes (mirrors the native-rack `SetTxRackBypass` gate and the RX EQ #59 mode gate).

• USER_GUIDE: new Phase Rotator section incl. a **"PHROT and wide / ESSB

audio (4–10 kHz)"** note (helps punch on comms-grade wide SSB; all-pass group delay smears hi-fi ESSB → many ops leave it off; A/B on own voice).

Release

• v0.4.0 already shipped, so this is **v0.4.1**. Bumped CMakeLists +

installer.iss, rebuilt clean, ISCC → `dist/Lyra-Setup-0.4.1.exe` (68.1 MB).

• Commits `6c3e8e7` (SWR/drive-cap/#172/docs) → `d99794d` (PHROT toggle) →

`165e176` (PHROT digital auto-off) → `eff9a5d` (version bump). Pushed `main` (`1c6de15..eff9a5d`), tag `v0.4.1`, GitHub release w/ installer.

• One snag: LNK1104 (operator had Lyra running) — asked operator to close,

relinked clean (the standing don't-force-kill-operator-Lyra rule).

2026-06-18 — CW TX ON AIR: paddle + keyboard console + CW-over-TCI (#105)

CW transmit went from "design mapped" to fully on-air this session.

CW-2 paddle (gateware iambic) — SHIPPED + operator-confirmed

• `applyCwKeyerEnable` arms `prn->cw.cw_enable` in CW mode (CWL/CWU) — the

firmware-keyer + sidetone master; paddle in the HL2 KEY jack keys the carrier autonomously (gateware CWTX), host stays out of timing.

• Carrier-on-marker fix: keyed carrier rides the marker in both sidebands

(single shared `cwPitchHz`); the off-marker was a DISPLAY bug (host MOX off the gateware key line switched to the TX analyzer) — fixed by NOT forwarding host MOX in firmware-keyer CW (QSK). Break-in selector QSK(default)/Semi/Manual; CW MON sidetone slider on the Audio panel.

• Manual break-in foot-switch confirmed ("tone footswitch and paddle"):

foot switch = PTT hold (gateware PTTTX), paddle keys within (→CWTX).

• Vestigial `cwSidetoneFreqHz` path removed (`14be1ed`) — CW pitch is the

single freq source.

CW-3a/3b host software keyer (CWX) + CW console — SHIPPED `8904749`

• Gateware-verified (radio.v / dsopenhpsdr1.v): host CWX keys the carrier

via the SAME CWTX path as the paddle — host drives `tx[0].cwx` (per element) + `cwx_ptt` (held per message); `cw_enable` (CW-mode armed) enables BOTH the host EP2 overlay AND the gateware CWX decode (`cmd_data[24]` = the 0x0f C1 bit). NO WDSP / no host carrier gen. The earlier "dot/dash bit" guess was WRONG — tracing corrected it.

• `tx/CwMorse` (table + PARIS timing + operator weight, pure/testable),

`tx/CwKeyer` (dedicated-thread element pump, monotonic absolute-deadline scheduling, interruptible abort), HL2Stream `sendCw/abortCw` + `setCwxKey/setCwxPtt` + lazy keyer + abort-on-close.

• `qml/CwConsolePanel` — chip→floating console (audio-rack idiom): WPM, type+Enter sends / Esc aborts, Send+Stop, reserved CW-5 decoder pane. "CW" launcher chip after the TX DSP strip.

• Design doc `docs/architecture/cw3_software_keyer_design.md`.

CW-4 CW-over-TCI — SHIPPED `fcfcb6b`, EESDR-spec-verified

• `tci_server` dispatch routes the EESDR TCI CW commands into the CW-3

keyer: `cw_macros` (plain send), `cw_msg` (contest prefix/call/suffix + `$N` repeat), `cw_macros_speed`/`cw_keyer_speed` (WPM, bidirectional), `cw_macros_stop`. Reserved-char un-escape `^~*`→`:,;`. sendCw/abortCw marshalled to the stream thread (QueuedConnection) — no race vs the console. CW-mode-only.

• **Verified vs the EESDR TCI protocol manual** (operator added

`docs/TCI Protocol.pdf` §3.2) + Thetis TCIServer.cs + SDRLogger+. The EESDR cross-check caught 3 drafting errors: `cw_macros_empty` is a server→client terminal signal (not a stop), reserved-char un-escape was missing, and `cw_msg` (the canonical contest cmd) was missing.

Docs

• `docs/help/USER_GUIDE.md` (`2a09824`): new "CW operating" section

(paddle/console/TCI + break-in + CW MON) + "CW keying over TCI" + TOC.

Standing directive (recorded)

• TCI work verifies against BOTH Thetis `TCIServer.cs` AND the EESDR TCI

manual (`docs/TCI Protocol.pdf`) — the authoritative spec.

Next

• **CW-5 (#173)** — RX CW decoder (faithful port of SDRLogger+'s

Bayesian/AFC/Farnsworth decoder, `hamlog/cw.html`) into the console's reserved pane + a macros field. Deep-read the decoder first.

• CW follow-ons: Semi/Manual host-MOX for CWX, F-key memories, CWFWKeyer

toggle; TCI `cw_terminal`/`cw_macros_delay`/`|..|`+<> macro syntax.

• On-air TCI CW bench via SDRLogger+ still TODO (spec-verified + builds).

Commit summary (today)

```
2a09824 docs(cw): USER_GUIDE — CW operating + CW-over-TCI
fcfcb6b feat(cw): CW-4 — CW keying over TCI, verified vs EESDR TCI spec
8904749 feat(cw): CW-3a/3b — host software keyer (CWX) + CW console
14be1ed refactor(cw): remove the vestigial cwSidetoneFreqHz path
```

2026-06-17 EOD — v0.4.0 RELEASED + main consolidated + CW TX mapped

UI readability batch (shipped `8fe2100`)

• TX DSP launcher chips (Speech/EQ/Combinator/Plating) → boxed buttons that

fill with the accent when their panel is open (were flat text-only).

• EQ graph: frequency-marker row across the top; x-range tracks the TX audio

passband (SSB/digital = high edge, AM/DSB/FM/SAM = half occupied) + amber dashed TX-edge marker; gain scale bigger (bold 11px) + brighter, labelled every 3 dB (added ±3/±9 rows), 0 dB bright blue.

• Speech/Combinator/Plate parameter labels + readouts brighter + bold 13px;

Combinator per-band micro-labels 9/10→11px. Operator: "easier on the old eyes." `.gitignore` += `_run_lyra_vac1.bat`, `Line`.

main consolidated (operator request)

• `origin/main` fast-forwarded `a3a517f..8fe2100` (clean, 32 commits, no foreign divergence). Local `main` moved up + checked out — `main` **is the working trunk again** (TX largely done; stops the "forgot to FF main" risk). `tx-rebuild` kept as a legacy ref.

v0.4.0 RELEASED

• Version bump 0.3.1→0.4.0 (CMakeLists + installer/lyra.iss); release notes

`docs/releases/v0.4.0.md`; commit `88f17f2`. Full rebuild (version baked), Inno Setup installer `dist/Lyra-Setup-0.4.0.exe` (~68 MB). Annotated tag `v0.4.0`, pushed main + tag, GitHub Release published with installer attached + verified. `main==origin/main==v0.4.0==88f17f2`.

#156 (restart-after-hard-kill) — PARKED

• Operator empirical: a real accidental hard kill of lyra.exe mid-VOICE-TX

(my pre-build force-kill fired while they were tuning) → **restarted CLEAN**. Only ever one observed hang (2026-06-13 mid-TUN). Intermittent → parked until it reproduces; prime suspect noted (infinite `hReadThreadInitSem.acquire()`), optional bounded-timeout hardening if revisited. PROCESS RULE logged: don't force-kill lyra for a build when the operator may be on air — ask first.

CW TX (#105) — DESIGN MAPPED (rainy-night research, no code)

• 2-lane reference dive (Thetis host cwx.cs/console.cs/TXA.c + HL2 wire

networkproto1.c + operator's HL2+ gateware RTL). Wrote

`docs/architecture/cw_tx_design.md` — full port plan.

• THE finding: HL2 CW is **100 % gateware-keyed** (no host/WDSP carrier;

`TXA.c` SetTXAMode has no CWL/CWU case; `radio.v:1145` bypasses EP2 TX-I/Q during CW, feeds the FPGA envelope ramp; FPGA also makes the hardware sidetone). Host just sends key-state. **The verbatim EP2 CW packer is ALREADY in tree** (`NetworkProto1.cpp:99-108`, dormant) + `prn->cw` seeded; EP6 dot/ptt decode already avoids the reference's left-shift bug. Missing: CW C&C; composer cases (0x0f/0x10/0x0b) + the native CWX keyer engine + FSM CW PttSource + UX. Build order C-1..C-5 in the doc. ■ flagged: 0x0b register collision with the shipped ATT-on-TX step-att — resolve before composing.

2026-06-17 (cont. — AM/DSB/FM native TX modulation + AM carrier #106/#93)

#165 follow-on → basic WDSP-native AM / DSB / FM transmit

Operator: keying AM showed only the upper sideband. RX was correct; the one-sided signal was the operator's own TX — the modulator was SSB-only.

• `hl2_stream.cpp` `setTxMode`: widened the WDSP-mode clamp `0..1` →

`0..13` so AM(6)/DSB(2)/FM(5)/SAM(10) reach `SetTXAMode` instead of being pinned to USB.

• `main.cpp` `wdspTxModeFor`: full mode map (LSB0…SAM10, default USB),

replacing the AM/FM/DSB/SAM→USB collapse.

• `main.cpp` `pushTxFilter`: per-mode sign-coded bandpass (TX mirror of

the RX §14.2 convention) — USB-side `+low..+high`, LSB-side `-high..-low`, double-sideband symmetric. NEVER calls `SetTXABandpassRun` (§15.23 trap); `SetTXABandpassFreqs` + `SetTXAMode` configure bp0. FM CTCSS silenced via `SetTXACTCSSRun(ch,0)` (TXA defaults it ON at 100 Hz).

• **TUN ■cw_pitch offset** generalized in `txDdsHzForTune` /

`txAnalyzerOffsetHz` so double-sideband modes stay centred.

• New wire bindings: `SetTXACTCSSRun`, `SetTXAAMCarrierLevel`

(`wdspcalls.{h,cpp}` — pointer + X-table + extern).

Doubled-bandwidth bug (operator bench) — FIXED

AM straddled centre correctly but occupied ~2× the set TX BW. Cause: the symmetric branch used `±high` = 2×high total. `txf->high` is the set TX BW (`Prefs.txBandwidth`); a double-sideband signal occupies the full ±span, so the edge must be **±high/2** — mirroring `WdspEngine::computePassband` (`half = bw/2`, ±half) which also draws the filter markers. Now AM at 6k = ±3k = 6k total, inside the markers. Operator bench: **correct**.

AM carrier level control (#93) — operator-facing, reference-faithful

• HL2Stream `amCarrierPct` Q_PROPERTY (+ getter/setter/signal/member,

QSettings `tx/amCarrierPct`), `TxControl.setAmCarrierLevel` callback, `main.cpp` lambda → `SetTXAAMCarrierLevel`. Settings → TX "AM Carrier" spinbox (0–100 %).

• **Mapping matches the reference exactly** (verified at

`setup.cs:9846`): UI is **% of standard carrier POWER**, coefficient = `√(pct/100) × 0.5`, **default 100 = standard AM** (= 25 % of PEP). So operator-stored values transfer 1:1. Helper `amPctToCarrierLevel()`, applied at both push sites. (Earlier draft passed pct→coeff straight, default 50 — corrected before bench.)

• N8SDR's exported profiles confirm the intent: `8K-AM-N8SDR`=40,

`*-N8SDR-AM`=84, `AM 10k CFC`=95 (reduced/controlled carrier).

Profile import — decided NOT to pursue (operator call)

DSP chain is Lyra-native (EQ/Speech/Combinator/Plate, not WDSP/CFC), so importing reference profiles is pointless — the *sound* lives in the native rack, which has no reference equivalent. `amCarrierPct` NOT added to the Profile struct. Profile manager stays as-is.

Status

All built clean + operator-bench-confirmed (AM/DSB/FM both sidebands inside the markers; AM carrier behaves). Committed this session; #106/#93 closed.

2026-06-17 (cont. — doc reconciliation + ATT-on-TX UI §15.31)

Doc reconciliation (operator-delegated while out)

• `THETIS_DIRECT_PORT_PLAN.md` reconciled (agent pass + my accuracy

corrections): obbuffs/P1 → [DONE]; compress/cfcomp/eqp → [N/A] (native-superseded); wcpagc leveler/ALC → [DONE] but **softened** — metering "shipped in code; #160 bench close-out + Brent/Timmy field-confirm still IN PROGRESS" (not over-claimed). PDF + DOCX regenerated via the generalized `tools/sync_execution_plan.py` (now `--md/--docx/--pdf-path/--title/--no-docx`). SESSION_LOG.pdf regenerated.

§15.31 — ATT-on-TX operator surface + visible lamp (operator-flagged)

Operator: "ATT on TX doesn't appear to be live — I don't see the LNA drop to –31 on TX." TRACED the path (did NOT re-assert from the doc — the [DONE] I'd let stand was operator-disproven):

• **WIRE path IS live + correct**: `fsmAdvance` raises

`setTxStepAttnDb(31)` on keydown (`hl2_stream.cpp:1651`) → `tx_step_attn=0` → XmitBit-gated `compose_case_11` C4 = `0x40` → ak4951v4 cmd_addr 0x0a rx_gain = 0 = min LNA during TX. The front end WAS being attenuated.

• **Root of the report = VISIBILITY gap** (operator confirmed "1 and 3"):

the LNA readout shows the RX setpoint (unchanged on TX); no on-screen confirmation, where Thetis visibly jumps the ATT to 31. Also the `src/tx/AttOnTxPolicy.{h,cpp}` class is **unused / wire-inert** — the live mechanism is inline in hl2_stream.cpp; the doc citation was wrong.

• **FIX** (operator-locked design confirmed before code; matches Thetis

Setup → General → Ant/Filters → "ATT on Tx" ✓ / "ATT: 31"):

  • HL2Stream: `attOnTxEnabled_` (default ON) + `attOnTxDb_` (default

31), Q_PROPERTY + setters + QSettings (`tx/attOnTx*`); `fsmAdvance` gates on the toggle + uses the value (replaces hardcoded `kAttOnTxDb`); mid-TX toggle re-applies live. Disabled → axis 0 (= reference `SetTxAttenData(0)`).

  • TxPanel.qml: "ATT" lamp in the gap between the Mic and Tune sliders, AUTO/TUN/MOX lit idiom — gray `ATT off` / orange `ATT 31` armed (RX) / solid-red + white-text `ATT -31` engaged (keyed). Engaged colour fixed from a salmon-on-maroon blend that read PINK → MOX-style solid red; label shows the negative (applied attenuation) on TX per operator request.

- settingsdialog.cpp: Settings → TX → "ATT on TX (RX-ADC protection)"

group (left/safety column) — Enable + ATT dB (0..31) spin, bidirectional with the lamp.

• Builds clean; operator-approved the UI in review. THETIS row kept

[WIP] (flips [DONE] only on the on-air bench confirm). AttOnTxPolicy class still unused (flagged for later delete-or-wire-up). Task #114 (broader: panadapter offset + PA-enable safety) stays pending.

2026-06-17 (build day — #162/#163 ship, #90 TX monitor 3-route, #164 Out picker; all HL2-confirmed)

Branch `tx-rebuild` (NOT pushed; `main`/origin stays v0.3.1). Every commit below was operator-bench-confirmed on the HL2+ before it landed.

#162 TX Speech profile NOT saving — FIXED (root cause CORRECTED vs the 06-16 theory)

The 06-16 "binding-break / UI-refresh" theory was **WRONG**. Operator pushback ("are we even capturing the Speech panel?") + a registry dump (`HKCU\Software\N8SDR\Lyra-cpp\profiles\item\`) proved it: the `speech` blob saved all-default. **Real root cause = QML signal shadowing** — the Stage card's custom `signal toggled(bool)` was shadowed by `AbstractButton.toggled`, so the ON button's `onClicked: toggled(checked)` emitted the Button's signal, never the model write → Speech always saved defaults.

• `c17960c`: qualify the emit `stage.toggled(checked)` → the model write

fires; registry-verified (`deessOn:true` … now persist).

• `46a428f`: `Binding{target:en; property:"checked"; value:on}` re-asserts

the toggle lamp from the model on recall (a checkable Button flips its own `checked` on click, severing the inline binding).

• Operator-confirmed: toggles save, recall re-lights lamps + restores

sliders, 2 fresh profiles round-tripped. Lesson: clean build + qmllint ≠ proof — verify by registry dump / launch.

#163 Rack-lamp dim in digital/CW + EQ analyzer recolor (cosmetic, operator-requested)

• `9be1fc1`: Speech/Combinator/Plate panels gray the ON lamp + dim controls

+ show an amber "bypassed (MODE)" header hint in DIGU/DIGL **and CW** (digital → `SetTxRackBypass`; CW → no mic). Purely visual off a `WdspEngine.mode` binding; model untouched, re-lights on USB/LSB.

• `558d5d9`: EqPanel — same dim + the operator's analyzer recolor:

Accumulate peak-hold → RED, pre-EQ "Before" overlay → WHITE (were near-white/amber), live "After" stays cyan. Line + RTA. Doc `66adc5e`.

#90 TX audio monitor — ALL THREE reference-faithful routes SHIPPED (#90 closed)

ONE shared post-rack tap (`xcmaster` case 1, READ-ONLY on `pcm->in`, after the rack / before `fexchange0`) → lock-free SPSC `MonitorRing` (`src/dsp/MonitorRing.h`); drained ONCE at the top of `dispatchAudioFrame` into `monScratch_`, shared by all routes. WdspEngine `monEnabled_`/ `monVolume_` (atomics, persisted); Audio-panel MON TX toggle + Monitor slider.

• Route 1 (HL2 jack, `6e6a282`): monitor mono on L/R in place of the

auto-muted RX when MOX up + MON on, via the existing `OutBound(0)` path.

• Routes 2+3 (`e1fd186`): Route 2 = VAC stream-2 + `SetIVACmon`/

`SetIVACmonVol` (ivacGet-gated, re-applied in `rebuildVac1`); Route 3 = TCI RX-audio (tap moved into `dispatchAudioFrame`; MON→monitor / TX→mute / RX→live).

• **Reference-VERIFIED (operator asked "is this Thetis or have we drifted?"):**

Thetis `audio.cs:386-404` drives `SetIVACmon(0,1)+SetIVACmonVol(0, monitor_volume)` on the PRIMARY VAC (id 0 = our `kVac1Id`) and `SetTCIRxAudioMox/Mon` (368-404) — the single MON drives jack + VAC + TCI, exactly as built. NOT drift.

• ■ Faithful behavior change: TCI RX-audio is now MUTED on the air (was

live RX during TX). Operator-bench-confirmed harmless — MSHV FT8-over-TCI decoded + completed a QSO (N8SDR↔KB6DAD to 73) with MON TX on, TX 5.1 W, DIGU rack auto-bypassed. Docs `266d8cd`.

#164 "Out" output-device picker wired + MON→MON TX relabel (`740f769`, doc `dc45485`)

The disabled placeholder "Out" button is now a live one-click output picker: a styled top-level popup (`popupType: Popup.Window` so it ESCAPES the short Audio-panel QQuickWidget clip — a plain Item-type popup was getting chopped at the panel bottom with no scroll). Lists HL2 jack + PC devices, current ■/cyan/bold, hover highlight, as-needed scrollbar. Reads the same `audioOutputDevices()`/`setAudioOutputDevice`/`audioDeviceIndex` the Settings → Audio combo uses; full config stays in Settings. MON button relabeled **MON TX** so its purpose is clear. ComboBox considered + rejected (long device names blow out the row width).

New items logged (not started)

• **#165 (■ bug):** AM/DSB modes only produce the UPPER sideband (lower

missing). AM/DSB are double-sideband → the WDSP passband for those modes is likely set one-sided (USB-only) instead of symmetric –W..+W. RX-DSP investigation, queued next.

• **GitHub / `main` branch:** operator has questions to raise on return.

Docs

• USER_GUIDE updated for the rack dim/recolor, the monitor (MON TX + the

VAC/TCI routes), and the Out picker.

• This session-log entry + a `THETIS_DIRECT_PORT_PLAN.md` reconciliation

pass (mark done-but-unmarked items, add a post-P4 shipped-work entry). `.pdf`/`.docx` of both NOT regenerated (no pandoc/doc-gen tool on this box) — MD sources updated; PDF/docx regen pending the operator's tool.

2026-06-16 (design + investigation day — NO commits; two items parked for tomorrow)

Context: native TX DSP rack (EQ/Speech/Combinator/Plate) + #49 Profile Manager bundle in flight; operator field-testing profiles on HL2+.

#90 TX audio monitor — DESIGN LOCKED, parked for tomorrow's build

Operator ask: MON button + Monitor volume slider on the Audio panel upper row. Studied Thetis MON / MonitorVolume (gates processed TX audio into the RX audio mixer + `SetIVACmon`; the verbatim `xMixAudio(0,0,…,out[2])` "mix monitor audio" line is already carried at `CMaster.cpp:458`, fed empty out[2]). Operator chose **both: HL2 jack now, separate PC device later**.

• **Key finding:** `WdspEngine::dispatchAudioFrame()` (wdsp_engine.cpp:3100)

STAYS LIVE during TX — RX audio is just gain-zeroed by the `txMuted_ && autoMuteOnTx_` gate, then `OutBound(0)` → EP2 LR → AK4951 jack. So the jack is live-but-silent during TX = the inject seam (no fight with RX stand-down).

• **Locked design — ONE shared tap:** in `xcmaster` case 1 capture

`pcm->in[stream]` AFTER the rack, BEFORE `fexchange0` (the processed "what you sound like" mic audio) → small SPSC ring.

• Route 1 (HL2 jack, NOW): drain ring in `dispatchAudioFrame`; when MOX

up + MON on, output monitor audio (scaled by new Monitor-vol) in place of muted RX audio → existing `OutBound(0)` path. No new stream.

• Route 2 (separate PC device, LATER): feed same tap into

`xvacOUT(kVac1Id, stream 2)` (replace `vacMonSilence_`) + `SetIVACmon` = **IVAC Stage 5 / DL-4** done reference-faithfully.

• Tradeoff (note in tooltip): post-rack tap is pre-WDSP-ALC/bandpass —

hear your rack DSP faithfully, not WDSP's corrective ALC.

• Operator: **"hold for tomorrow's build."**

#162 Speech-panel profile not saving — ROOT-CAUSED, parked first-thing tomorrow

Operator: TX Speech panel options aren't saved/recalled by Profiles while EQ/Combinator/Plate are. Decisive repro: turned De-esser ON, recalled an all-off profile → **De-esser button stayed ON.**

• Traced the full path: capture (main.cpp:752) / apply (main.cpp:798) /

`Profile` toJson+fromJson+sameValues / `SpeechModel` saveState+loadState / single instance / context property — **all symmetric & correct.** The data DOES save and DOES apply to the model + DSP.

• **Root cause = QML binding-break (UI-refresh, not persistence):**

SpeechPanel (and PlatePanel/CombinatorPanel) use `Button{checkable:true; checked:}` + `Slider{value:}`. The first click/drag makes Qt Quick write `checked`/`value` imperatively, **severing the binding to the model.** Recall then updates model+DSP+numeric labels but NOT the toggle/slider visuals — the button is a dead light. EqPanel dodges this (non-checkable toggles + pure `checked:` binding off a `rev` counter + `Q_INVOKABLE` getters). De-esser is the clearest demonstrator because a stuck binary toggle is unambiguous; Plate/Combinator share the latent pattern but only surface after mid-session interact-then-recall.

• **Fix (#162, high pri):** adopt EqPanel's binding-safe pattern in the

three rack panels. QML-only, no DSP/wire risk.

Tomorrow's order (operator-set)

1. **#162** Speech (+Plate/Combinator) panel binding-safe rebuild — FIRST.

2. **#90** TX monitor build (Route 1 HL2 jack; Route 2 PC device as the

follow-on within #90 / IVAC Stage 5).

Memory: `project-lyra-cpp-monitor` (the #90 locked design).

2026-06-13 (Saturday — ■ TX BRING-UP COMPLETE + v0.2.3 RELEASED)

`main` **==** `tx-rebuild` **== HEAD** `e8aafbc` **(pushed).** Tags: milestone `tx-working-2026-06-13` (at `bd61d07`) + release `v0.2.3` (at `a5dc763`). GitHub Release **v0.2.3 — "Transmit: SSB + digital (TCI) on the air"** PUBLISHED with `Lyra-Setup-0.2.3.exe` (67.7 MB) attached, marked Latest.

P4.b Wire-LIVE switchover — SHIPPED + operator HL2 bench PASSED

• `fb9ec41` P4.b-1 — Wire-LIVE RX-out gate (operator bench PASSED).

• `0b551b4` P4.b-2 — TX wire-LIVE: first modulated RF through the

direct-port chain (operator HL2 bench).

• `9db2c50` P4.b-2.1 — TX SSB sideband fix: USB transmits USB (9100-verified).

• `84dbb12` fix(tx): TUN panadapter spike on-marker — TX-analyzer crop

shifted by NCO–dial offset. Verified Lyra TUN is **Thetis-exact zero-beat** (two-offset scheme; the bug was display-only — authoritative refs: console.cs:32553-32587 UpdateTXDDSFreq path, postgen 30788-30800, readout 22035-22051).

• §7 dead-code retirements (3 build-gated commits): `fe6a438`

OutboundRing + Ep2SendThread, `99fd24a` txWorkerLoop + composeCC + buildEp2KeepaliveTemplate, `e711256` old HL2-audio-jack EP2 ring (pushAudio + setHl2AudioSink).

• P4.b bench gates PASSED: SSB voice both sidebands + Phase-3-EXIT

RF-safety kill-test (RFG drops on hard-kill).

TCI digital TX re-home — SHIPPED + ON-AIR VALIDATED

• `bd61d07` — filled the §10.3 `src/tci/TciTxBridge.{h,cpp}` skeleton

(singleton, mono deque, I=Q=mono drain w/ zero-fill underrun, `queuedSamples()` for CHRONO). Registered via `SendpInboundTCITxAudio(&TciTxBridge;::inboundCb)` after create_cmaster; fed by `tci_server.cpp` handleBinaryFrame tap; gated by `SetTXTCIAudio(0, micSource=="tci")`. The verbatim DSP seam (xcmaster case 1 use_tci_audio gate) was already present — only the host-side sink was missing.

• **Bench: MSHV → TCI → FT8 real on-air QSOs** — N3YDN (FM29),

F4FLF (JN18, ~4160 mi DX), KO4OIG (EL89), W3EWL/QRP.

Space-bar PTT toggle (#157) — SHIPPED

• `2824ebf` — `Prefs.spaceBarPttEnabled` (tx/space_bar_ptt_enabled,

default ON), MainWindow keydown/keyup gate, Settings → Hardware → Transmit checkbox under the HW-PTT tickbox. Operator-verified working.

Release

• `e8aafbc` — version bump 0.2.2 → 0.2.3 (CMakeLists project(VERSION) +

installer/lyra.iss AppVersion). Clean build, ISCC installer, tag, GitHub release (notes + installer).

NEXT (morning)

• **Mic-input paths discussion** (PC/VAC host audio + plain-HL2 vs HL2+).

Key reframe: a **plain HL2 (no AK4951) has no radio-side mic at all** → host audio-in (#102 VAC1) is REQUIRED for those ops to TX phone, not optional. All sources are interchangeable 48 k mono feeders of the same `InboundTCITxAudio` seam (TciTxBridge = the template). §5.4 "single mic-source class, no abstraction" rule has expired now that there are 4 concretes — introduce `ITxAudioSource`. Capability-aware default (plain HL2 → PC/VAC; HL2+ → AK4951 mic). Decide: TX-in-only vs full bidirectional VAC; Qt Multimedia vs WDSP rmatchV.

• Parked: #156 restart-after-hard-kill-mid-TX hang; intermittent

menu-drag display stutter (monitoring; ruled out as a §7 regression);

#104 HL2 Line In (codec mux); #103 VAC2.

2026-06-12 (Friday PM-4 — P4.a prep SHIPPED, both commits wire-inert)

Branch: `tx-rebuild`, HEAD `baef866` (pushed).

P4.a-1 SHIPPED `a4aa8c3` — sendProtocol1Samples verbatim (DORMANT)

- NEW `src/wire/NetworkProto1.cpp` — networkproto1.c PARTIAL: the EP2 writer thread `sendProtocol1Samples` verbatim (networkproto1.c:1204-1267). Mechanical diff IDENTICAL (whitespace-normalized, accommodations mapped back); sole structural delta = braces around the DEFERRED non-HL2 else-branch (comment-only branch needs braces to compile). Accommodations documented in the preamble: lyra::wire namespace; the HPSDRModel enum-class mapping; `write_main_loop_hl2(prn->OutBufp)` = the FrameComposer monolithic WriteMainLoop_HL2 equivalent (#121/#122 fold); generic WriteMainLoop (non-HL2/ANAN P1) carried as DEFERRED reference text; (AVRT_PRIORITY) cast + suppress 4100 on the verbatim signature.
- `io_keep_running` global added (network.h:411 verbatim; init 0 so the loop body cannot run even if started prematurely) + the decl in RadioNet.h.
- FrameComposer `write_main_loop_hl2` tail flipped from the Step-14 `outbound_notify_consumed_pair()` translation to the verbatim `ReleaseSemaphore(prn->hobbuffsRun[0/1], 1, 0)` pair. Dormant-safe: nothing calls write_main_loop_hl2 until P4.b starts the writer; the hobbuffsRun handles are created at the P4.b open() with the quartet.
- DORMANT: nothing `_beginthreadex`'s the function until P4.b's open() (reference start site = StartAudio, netInterface.c:66 + the quartet :68-71). Build clean, zero new warnings.

P4.a-2 SHIPPED `baef866` — PostGen seam entries + verbatim field types

- **wdspcalls**: SetTXAPostGen{Run,Mode,ToneMag,ToneFreq,TTMag,TTFreq} for the P4.b TUN re-home (legacy DC-injection TUN dies with the legacy EP2 packer; TUN becomes the reference TXA output-side tone generator). Pre-cdef audit: NO header declares these — PORT-exported definition sites in wdsp/gen.c only (gen.c:784/792/800/808/817/826), signatures harvested from the definition bodies; presence + exact casing verified against the bundled wdsp.dll export table (PE scan, all 6 PRESENT of 533). TT pair included per table rule #3 (PS committed; the PS calibration drive is the two-tone).
- **TxReadBufp**: std::vector -> verbatim `double*` (network.h:62) + the verbatim calloc at create_rnet (netInterface.c:1604, 2*sizeof(double)*720, process lifetime — the P2.b OutBufp precedent). Ep6RecvThread's 5 `.data()` sites -> bare pointer; the

vector-size guard reduces to a null check (fixed 1440-double capacity >= 4*spr for every valid nddc).

- **hWriteThreadMain**: std::thread -> verbatim HANDLE (network.h:90). Zero users existed on the std::thread form; P4.b assigns the verbatim `_beginthreadex(sendProtocol1Samples)` handle.
- Build clean (sole warning = the pre-existing hl2_stream C4456, untouched file). Commit-message nit: a backtick-quoted literal got eaten by bash substitution again (the P2.b class) — citation survives, no force-push.

NEXT = P4.b (THE switchover — ONE commit, full operator HL2 bench gate)

- open(): quartet CreateSemaphore ×4 + `prn->hWriteThreadMain = _beginthreadex(sendProtocol1Samples)` (io_keep_running=1 BEFORE start).
- dispatchAudioFrame -> the asioOUT-pattern tee (HL2-jack = real audio into OutBound(0); PC = zeroed tee so EP2 pacing never starves) — the B.6.b-delicate RX switchover.
- `Inbound(inid(1,0), mic_sample_count, prn->TxReadBufp)` at the live EP6 mic-harvest site (Ep6RecvThread, the block already mirrors networkproto1.c:560-579).
- FSM re-home: TXA SetChannelState arming per the §15.25 ground truth + TUN -> SetTXAPostGen.
- Control-plane mapping per the design doc §5 table; retirements per §7 (txWorkerLoop + keepalive + DC-TUN + slew-fill, OutboundRing, Ep2SendThread, FrameComposer cv tail).
- Teardown: io_keep_running=0 -> release both hsend sems once -> join -> CloseHandle quartet -> destroy_obbuffs BEFORE destroy_xmtr.
- Bench gate per design doc §8 incl. explicit TX-state lines + FIRST VOICE through the chain.

2026-06-12 (Friday PM-3 — P2 audit complete + P2.a eer completion SHIPPED)

Branch: `tx-rebuild`, on top of `2897756`.

P2 fidelity audit (the deliverable that scopes P2.b/c/P4)

- **Reference TX-out chain (HL2 P1), fully read:** `ob_main` -> `sendOutbound` (network.c:1237; P1 branch :1285-1340) — id 1 memcpys 126 complex into `prn->outIQbufp` + Release(hsendIQSem) + Wait(hobbuffsRun[0]); id 0 -> `outLRbufp` + hsendLRSem + hobbuffsRun[1] -> `sendProtocol1Samples` thread (networkproto1.c:1204-1267): WaitForMultipleObjects(both, TRUE) -> eer overwrite if `peer->run && XmitBit` -> `!XmitBit => memset outIQbufp` -> swap_audio_channels -> 16-bit BE quantize + HL2 CW-bit overlay into `OutBuf` -> WriteMainLoop_HL2 -> MetisWriteFrame -> Release both hobbuffsRun. Resources: bufs calloc'd netInterface.c:1606-1608; semaphore quartet created in StartAudio :68-71; obbuffs rings 0/1 created in UpdateRadioProtocolSampleSize :1856-1857; eer created run=0 per xmtr (cmaster.c:212-224).
- **Lyra dormant Step-14 surface:** functionally-parallel idiom translations predating the verbatim mandate — prn buffers are std::vector (RadioNet.h:552-554), the semaphore quartet became OutboundRing's bool+cv, Ep2SendThread is the sendProtocol1Samples rewrite. `XmitBit` is already verbatim (extern int, RadioNet.h:733). Disposition: re-port verbatim, retire the translations at P4 Wire-LIVE.
- **Landmine found:** BOTH reference functions deref `pcm->xmtr[0].peer->run`, but P0.d had deferred the eer creation — `peer` was NULL. That made P2.a unambiguous and first.

P2.a SHIPPED — eer completion (verbatim)

- `wire/cmcomm.h`: opaque `eer` typedef replaced by the FULL verbatim struct (wdsp/eer.h:30-49, mechanical diff IDENTICAL); opaque `DELAY` twin typedef added (wdsp/delay.h:32-59).
- `wire/wdspcalls.{h,cpp}`: +5 seam entries — create_eer / destroy_eer / xeer / pSetEERSize / pSetEERSamplerate (signatures verbatim from eer.h:51-75; all five verified present in the bundled wdsp.dll via PE export-table scan).
- `wire/CMaster.cpp`: all four deferred eer sites RESTORED verbatim — create_xmtr `create_eer(run=0, ...)` 12-arg block (cmaster.c:212-224); destroy_xmtr `destroy_eer` (:262, after destroy_ilv per reference order); xcmaster `xeer(pcm->xmtr[tx].peer)` (no-op at run=0 with in==out — verified in eer.c:86-122); SetXmtrChannelOutrate pSetEERSamplerate/pSetEERSize. Restored-lines verbatim-subset check PASS (18 lines).
- Runtime impact: create_xmtr now also creates/destroys one run=0 eer object per xmtr (heap + CS only); xeer in the TX pump body never executes today (stream-1 pump has no producer). RX path untouched. Clean build, zero warnings.

P2.b SHIPPED (same session) — prn outbound surface verbatim

- `outLRbufp`/`outIQbufp`/`OutBufp` std::vector -> the VERBATIM reference pointer fields (network.h:64-66) + the verbatim calloc set at create_rnet (netInterface.c:1606-1608, mechanical subset PASS). Lifetime = process (create_rnet call-once) = reference.
- Verbatim HANDLE quartet declared on prn (hsendLRSem/hsendIQSem/hsendEventHandles[2]/hobbuffsRun[2], network.h:92-95) — DORMANT/nullptr until P4's StartAudio equivalent; the Step-14 cv+flags translation stays alongside until P4 retires its consumers.
- Callers bent: OutboundRing `.data()` ×6 -> raw; Ep2SendThread pack casts -> char; `write_main_loop_hl2` parameter -> `const char*` (reference type). Wire-inert (only create_rnet's allocation runs live). The lone C4456 in hl2_stream.cpp:776 is PRE-EXISTING (file untouched; surfaced by the header-triggered recompile) — flagged for a separate cleanup.

P2.c SHIPPED (same session) — sendOutbound verbatim + ob_main restore

- NEW `src/wire/Network.cpp` — network.c PARTIAL direct port: `sendOutbound` verbatim (network.c:1237-1341). P1 branch LIVE (the outLRbufp/outIQbufp memcpy + hsendLRSem/hsendIQSem release + hobbuffsRun wait handshake, incl. the EER de-interleave sub-branch with its function-local static ptr); ETH branch carried as DEFERRED reference text (Protocol 2/ANAN = v0.4; WriteUDPFrame/udpOUT unported). Accommodations, each documented inline: the RadioNet.h enum-class mapping on the protocol gate; suppress 4101 (ETH-only locals) + 4456 (the reference's own loop-scope `i` shadowing its function-scope `i`, network.c:1239 vs :1298). Mechanical diff vs network.c:1237-1341 = IDENTICAL (whitespace-normalized, deferred text restored).
- `wire/ObBuffs.cpp` ob_main: the DEFERRED `sendOutbound(id, a->out)` hand-off RESTORED verbatim — the P1 TU is now reference-complete.
- Still dormant end-to-end: no create_obbuffs caller until P3/P4; the semaphore quartet is created by P4's StartAudio equivalent before the pump can deliver, the same ordering the reference relies on. Clean build, zero warnings.

P3 SHIPPED (same session; survived + re-verified after an operator power outage mid-arc) — netInterface registrations + obbuffs ring lifecycle

- **create_rnet moved to once-per-process** at the main.cpp QTimer block AFTER create_xmtr — the reference's C#-init ordering (create_cmaster -> create_xmtr -> create_rnet -> StartAudio). The old per-open call re-allocated prn on EVERY stop/start (leak + wire-state reset, contradicting the close() "prn stays alive for re-open" contract — a latent pre-existing defect the P3 reference-read surfaced). open() now asserts the reference prn-non-null contract (netInterface.c:40); if wdsp.dll failed to load, open() refuses with a qCritical (no WDSP = no radio, the reference posture).
- **SendpOutboundTx(OutBound) restored at the reference site** (netInterface.c:1761, tail of create_rnet) — pcm->OutboundTx -> xmtr[0].pilv via the verbatim no-guard SetILVOutputPointer; ordering safe by construction. The Stage C.3-fix "registration belongs at the xmtr-open site" note SUPERSEDED (it guarded a Step-14 stub against per-open clobber; under the direct port the reference site is correct). RX side deliberately NOT registered — WdspEngine's openRx1 owns RX dispatch in the approved hybrid until P4 (registering RX here = the B.6.b clobber = dead RX audio).
- **UpdateRadioProtocolSampleSize ported verbatim** (netInterface.c:1836-1858, mechanical diff IDENTICAL with the documented kMax*/enum-class accommodations) — called per session-open from HL2Stream::open() at the StartAudio position (:45): per-protocol spp values (USB: mic 63 / rx 63 / tx 126 / audio 126) + create_obbuffs(0/1) — TWO new ob_main pump threads per session, both idle (ring 0 has no producer until P4's RX switchover; ring 1's producer is xilv in the stream-1 cm pump, quiescent until P4 feeds Inbound(1,...)). destroy_obbuffs(0/1) at close() per StopAudio (:112-113).
- Power-outage recheck: all 4 edited files probe-verified intact, diff-vs-HEAD = exactly the P3 hunks, forced rebuild of the touched TUs clean (sole warning = the pre-existing hl2_stream C4456), mechanical diffs re-PASSED post-outage.
- [DONE] **OPERATOR HL2 BENCH PASSED (same day)**: "rx fine" — RX working on the P3 lifecycle (once-per-process create_rnet, per-session obbuffs rings + 2 idle pump threads, TX registration live). **P3 gate cleared; P4 Wire-LIVE unblocked.**

■ Operator-flagged near-miss + bench-discipline correction (2026-06-12)

- Operator accidentally pressed TUN a few sessions back and got REAL RF OUT, despite the rebuild-arc bench notes repeatedly saying "TX stays wire-quiescent / RX-only." Code-verified: those statements were true of the NEW direct-port chain only. The OLD hardware-validated TX-0c path — TUN DC-injection at the EP2 packer (hl2_stream.cpp:~2516) + the requestMox FSM + ATT-on-TX — was NEVER part of the TX rip (only the SSB voice DSP subsystem was removed) and has been LIVE on tx-rebuild the whole time. RF required the operator's own persisted PA-enable opt-in + drive settings (the deliberate-arm gate worked as designed; it was armed from the first-RF benches and persisted).

- **Operator decision: LEAVE IT** — it is the proven path, gated behind the PA-enable opt-in; P4 replaces it under its own full bench. (Disarm anytime: un-tick Enable PA in Settings.)
- **Discipline correction going forward:** every direct-port bench gate now includes an explicit "TX state unchanged" line (TUN/MOX behavior + PA-enable posture), and "RX-only / wire-quiescent" is reserved for statements about the WHOLE wire, not just the new chain. Applies to the P4 gate and onward.

P2 COMPLETE. NEXT

1. **P3** — netInterface outbound registrations: `SendpOutboundRx(OutBound)` / `SendpOutboundTx(OutBound)` per netInterface.c:1749-1761 — MUST register AFTER create_xmtr (verbatim setters have NO null guards); plus the create_obbuffs(0/1) sites (netInterface.c:1856-1857) at the Lyra equivalent of UpdateRadioProtocolSampleSize.
2. **P4** — Wire-LIVE switchover (one commit, full HL2 bench gate): verbatim sendProtocol1Samples thread + the semaphore-quartet creation + WriteMainLoop_HL2 hand-off; retires the OutboundRing/Ep2SendThread cv translations.

2026-06-12 (Friday PM — P1 obbuffs.c verbatim direct port SHIPPED)

Branch: `tx-rebuild`, on top of `976062d` (the P0.d bench-PASS doc flip).

Shipped (one commit)

- **NEW** `src/wire/ObBuffs.{h,cpp}` — reference obbuffs.{h,c} verbatim: `obb`, `*OBB` twin typedef + numRings/obMAXSIZE/OBB_MULT defines + the file-scope `_obpointers obp` four-alias bank + create/destroy/flush/OutBound/obdata/ob_main/SetOBRingOutsize. The TX-OUT seam (WDSP output → ring → ob_main pump thread → sendOutbound → protocol packers); SEPARATE TU from cmbuffs (inbound-only).
- The reference's own **2014-era idioms kept as shipped** (deliberately NOT harmonized to the cmbuffs sibling): calloc/free (not malloc0/_aligned_free), destroy-time `obp.pcbuff[0]==NULL` guard, obdata WITHOUT the MW0LGE CS wrap, the csOUT enter/leave pair at the top of the ob_main loop, the `out` work buffer inside the struct.
- **Sole deferred line:** `sendOutbound(id, a->out);` in ob_main — the reference defines it at network.c:1237 (the P1 branch :1285-1340 is the outIQbufp/outLRbufp + hsendIQSem/hsendLRSem/hobbuffsRun handshake WriteMainLoop_HL2 consumes) = **P2 scope**. Carried in place as reference text with a DEFERRED tag; decl in ObBuffs.h verbatim.
- Provenance check resolved before code: the "OutBound at network.c:1285-1340" citation in older RadioNet.cpp comments is the PROTOCOL_1 **branch of sendOutbound**, not a second OutBound — obbuffs.c's OutBound is the one registered via SendpOutboundRx/Tx (P3).
- Compiler accommodations: same set as CmBuffs.cpp (suppress 4312/4189/4311/4302 on the thread-arg casts + `(AVRT_PRIORITY) 2`), each marked inline.

Verification

- Clean build, ZERO warnings.
- Mechanical diff vs reference: **ObBuffs.cpp IDENTICAL** (code lines, whitespace-normalized); ObBuffs.h sole delta = the reference's include-guard `#define _obbuffs_h`, replaced by `#pragma once` (documented packaging difference, same as CmBuffs.h). Live-line-subset PASS on both.

Wire impact

- **NONE — the TU is dormant.** No Lyra caller of create_obbuffs exists until P3/P4 (the reference creates rings 0/1 in netInterface.c:1856-1857). No new threads at startup, wire bytes unchanged. An operator RX smoke at convenience is prudent (binary changed) but this touches no live path.

NEXT

1. **P2 = sendOutbound / sendProtocol1Samples fidelity audit** of the dormant wire layer (reconcile network.c:1237-1341 + networkproto1.c sendProtocol1Samples vs Lyra's Step-14-era OutboundRing/MetisFrame/Ep2SendThread surface), then restore the ob_main hand-off verbatim.
2. P3 netInterface registration (SendpOutboundRx/Tx(OutBound) — AFTER create_xmtr, the verbatim setters have NO null guards) → P4 Wire-LIVE (one commit, full HL2 bench gate).

2026-06-12 (Friday — P0.d CmBuffs/CMaster/cmsetup verbatim direct port SHIPPED + ■ HL2 RX-regression bench PASSED)

Branch: `tx-rebuild` HEAD = `afc7950`, pushed to `origin/tx-rebuild`.

■ **OPERATOR HL2 BENCH (same day): RX-regression gate PASSED** ("RX still working") — clean RX on the new per-stream cmbuffs pump/ring layout (two cm_main threads, stream 0 idle by design). P0.d is fully closed; **P1 (obbuffs.c verbatim port) is UNBLOCKED** and starts next session.

Shipped (one commit, `afc7950`)

- **NEW** `src/wire/cmsetup.{h,cpp}` — reference cmsetup.{h,c} verbatim: cmMAX* sizing macros (16/4/4/2/32), rxid/txid/sp0id/stype/chid/inid/mixinid/getbuffsize, SetRadioStructure + set_cmdefault_rates. CreateRadio/DestroyRadio carried with the unported pipe/sync calls commented.
- `src/wire/CmBuffs.{h,cpp}` **rewritten verbatim** — `cmb`, `*CMB` twin typedef, `#define CMB_MULT (3)`, malloc0/_aligned_free, no calloc/intptr_t/guard deviations; pcm->in[] allocation moved back to create_cmaster (reference shape).
- `src/wire/CMaster.{h,cpp}` **rewritten verbatim** — FULL `_cmaster` struct (cmaster.h:39-99, PS surface incl. out[3]/panalalloc/pgain/peer), `cmaster`, `*CMASTER`, raw TCI fn ptrs, `enum AudioCODEC`, `cm = {0}`. **TxChannel RAII carve-out DELETED** (src/wdsp/TxChannel.{h,cpp} retired): verbatim no-arg create_xmtr opens the WDSP TXA channel (chid(1,0)=1) + TX analyzer (disp 1; pan analyzer = disp 0, no collision) + out[0..2] + create_ilv itself through the wdspcalls seam. create_cmaster/destroy_cmaster verbatim per-stream loops (update[] CS + cmbuffs + in[] for all cmSTREAM streams; create_rcvr = deferred stub, RX hybrid). xcmaster verbatim: update[] critical section restored, real stype/txid/chid, TCI-override memset restored; fexchange0 + monitor-mix xMixAudio (accept-gated, quiescent vs the 1-input WdspEngine mixer) + xilv live. SetXcmInrate/SetCMAudioOutrate/SetRcvr-/SetXmtrChannelOutrate/SetRunPanadapter/SetAntiVOXSource* ported. All deferred subsystem lines carried IN PLACE as reference text with DEFERRED tags.
- `src/wire/cmcomm.h` — opaque verbatim twin typedefs ANB/NOB/EER/VOX/TXGAIN/ANALYZERS (tags match reference headers; completing a type later is source-compatible) + the umbrella-mapping note (.cpp files include explicit family headers — an include-list umbrella would be circular under #pragma once because the family headers include cmcomm.h for the base surface).
- **main.cpp** — SetRadioStructure(2,1,1,1,0,…)/set_cmdefault_rates(48 k) config block BEFORE create_cmaster (derived ids match the live layout: chid(0,0)=0 = WdspEngine RX1 channel, chid(1,0)=1 = TX channel); create_xmtr() invoked in the QTimer block after resolve_wdsp_calls (the documented DEFERRED-CALLSITE accommodation); handler-1.5 = gated destroy_xmtr(). Consumer-side decls for the headerless PORT functions (test_ilv precedent). RadioNet.cpp → enum AudioCODEC cases; scratch/test_ilv.cpp → verbatim `cmaster cm = {0};` globals.

Verification

- Clean build, ZERO warnings (C4701 in getChannelOutputRate disabled function-wide with a documented verbatim-text-wins pragma — line-level suppress can't reach the code-gen-stage warning).
- scratch/test_ilv.exe ALL PASS against the new verbatim globals.
- Mechanical diff vs reference: CmBuffs.{h,cpp} / cmsetup.h / CMaster.h struct+decls+enum / all 10 fully-verbatim cmaster.c functions IDENTICAL (whitespace-normalized); cmsetup.cpp comment-only deltas; live-line-subset check PASS for the partially-deferred bodies. Sole code accommodation: `(char *)""` at the XCreateAnalyzer call (C++ string-literal constness).

Behavior changes at startup (for the bench)

- TWO cm_main pump threads now start (streams 0 + 1; stream 0 idles forever — no Inbound producer; the reference layout). Stream rings sized from cmMAXInRate=384000 (in[] = 512 complex).
- TX stays wire-quiescent: no Inbound() producer yet, pcm->OutboundTx null until P3.
- destroy ordering: handler-1.5 destroy_xmtr() (gated) → handler-4 destroy_cmaster() = reference relative order.

NEXT

1. ~~**Operator HL2 RX-regression bench** (the P0.d gate)~~ — ■ **PASSED 2026-06-12** (RX working; gate cleared).
2. **P1 = obbuffs.c port** (TX-out seam, separate TU from cmbuffs) → P2 sendOutbound audit → P3 netInterface registration (outbounds AFTER create_xmtr) → P4 Wire-LIVE (one commit, HL2 bench gate).

2026-06-08 (Monday EOD — STAGE B aamix.c port COMPLETE + bench-validated)

Branch: `tx-rebuild` HEAD = `533b06b`. Pushed to `origin/tx-rebuild` (0/0 in sync). `origin/main` deliberately untouched (different arc). **Backup:** `_backups/lyra-cpp-2026-06-08-aamix-stage-B-COMPLETE.bundle` (16 MB, `git bundle --all`).

Today's shipped commits (in order, all on `tx-rebuild`)

```
533b06b [aamix port -- Stage B.6.b-final]   strip diagnostic instrumentation
8b8e0da [aamix port -- Stage B.6.b-fix1]    remove Stage-A NO-OP SendpOutboundRx stub (THE FIX)
12369bc [aamix port -- Stage B.6.b-debug-sink] dispatchAudioFrame sink-side counters
225518c [aamix port -- Stage B.6.b-debug]   9-counter chain instrumentation
53d9e5a [aamix port -- Stage B.6.b-retry]   wire AAMix via reference-faithful path
6e773a5 Revert "[aamix port -- Stage B.6.b] wire AAMix(1-input passthrough) into RX path"
525c2f0 [aamix port -- Stage B.6.b]         wire AAMix(1-input passthrough) -- REVERTED at bench
27ac2fa [aamix port -- Stage B.6.a]         extract feedIq audio tail into dispatchAudioFrame
```

Arc summary

- **B.0 → B.5:** AAMix.h/cpp port complete from earlier sessions (538-line header verbatim from `aamix.c` with GPL v3+ attribution to NR0V/WDSP, 1100+ lines of cpp; idiom translations from Win32 to C++23: `CRITICAL_SECTION→std::mutex`, `HANDLE semaphore→std::counting_semaphore`, `_beginthread→std::jthread`, `_Interlocked*→std::atomic` with fetch_or/fetch_and, `malloc0→std::vector` RAII).
- **B.6.a (`27ac2fa`):** Refactored `feedIq` audio tail (lines 2402-2456) into new `WdspEngine::dispatchAudioFrame(audio, nframes)` helper. Byte-identical relocation; bench-confirmed.
- **B.6.b first attempt (`525c2f0`):** Wired `aaMix_` into RX path with `create_aamix(active=0x01L)` shortcut. **Operator bench: NO audio.** Reverted (`6e773a5`).
- **B.6.b-retry (`53d9e5a`):** Reference-faithful per `cmaster.c:297-313`: `create_aamix(active=0L, ring_size=4096, slew=0/10/0/10ms)` → `SetAAudioMixOutputPointer(aaMix_, 0, outbound)` → `SetAAudioMixState(aaMix_, 0, 0, 1)` (activate via close_mixer/open_mixer slew atom). **Operator bench: STILL NO audio.**
- **B.6.b-debug (`225518c`):** Chain-side instrumentation — 9 atomic counters + 1 ENTRY beacon + producer/consumer 1-Hz log emits in `xMixAudio` / `mix_main`. **Operator bench result:** chain healthy end-to-end (`xmix=rel=wake=out=188/sec`, `nzOut=99%`, `mixmain_started=1`, `Outbound=set`) but operator confirmed STILL no audio. ⇒ defect is downstream of Outbound.
- **B.6.b-debug-sink (`12369bc`):** Dispatch-side instrumentation — 6 atomic counters in `dispatchAudioFrame` + 1-Hz log line tag `[disp-dbg]`. **Operator bench result:** `[disp-dbg]` lines NEVER fired in the entire 33-second capture. ⇒ `dispatchAudioFrame` is never entered. ⇒ the Outbound callable AAMix is calling is NOT the lambda WdspEngine wired.
- **B.6.b-fix1 (`8b8e0da`, THE FIX):** Root cause traced from bench timestamps:
  - `19:22:02.972` — `WdspEngine::openRx1` calls `create_aamix(id=0, ..., Outbound=dispatchAudioFrame_lambda)` ⇒ `paamix[0]->Outbound = real lambda` ✓
  - `19:22:02.973` — `HL2Stream::open() → create_rnet() → SendpOutboundRx(STUB)` ⇒ `pcm->OutboundRx = STUB`, then `SetAAudioMixOutputPointer(nullptr, 0, STUB)` resolves `nullptr → paamix[0]` ⇒ `aaMix_->Outbound = STUB`**, real lambda CLOBBERED** ✗
  - The Stage-A NO-OP stub at `RadioNet.cpp:272-283` was a placeholder ("Stage B aamix port wires it via SetAAudioMixOutputPointer") for the wire-up Stage B was supposed to replace. Stage B.6.b correctly wired `dispatchAudioFrame` in `openRx1` — but left the Stage-A stub registration in place, which then overwrote the wiring 1 ms later when `HL2Stream::open()` called `create_rnet()`.
  - **Fix:** delete the Stage-A stub registration. Comment block on the deleted case warns future hands that the wire-up needs the per-channel sink-routing context (`hl2Out_`, `hl2AudioPush_`, `audioRing_`) that the global `pcm` cannot reach — so the wire-up must live at `openRx1`, not `create_rnet`.
- **B.6.b-final (`533b06b`):** Strip diagnostic scaffolding. -134 lines. Brief comment in `AAMix.cpp` points future hands at commit history (`git show 225518c` for chain instrument, `git show 12369bc` for dispatch instrument).

Operator HL2+ bench (final, 19:27:49 → 19:28:47 — 58-second run)

- HL2 jack startup: first `[disp-dbg]` post-unmute: `calls=188 muted=0 gain=0.0845 peak16=32767 hl2(push/null)=188/0 hl2pushSet=1 audioRingSet=0` — full-scale audio reaching AK4951.
- Operator-confirmed audible RX ("Yes we have audio now").

- Mute toggle via dispatch (muted counter 0→22→133→287).
- Output device switched to PC Soundcard at 19:28:17 → `hl2Out=0, audioRingSet=1, pc(push) 176→928/sec`.
- Switched back to HL2 jack at 19:28:23 → `hl2Out=1`, `hl2(push)` resumed climbing.
- AGC mode changes (med/slow/off/fast) clean.
- Chain `out=N` tracks 1:1 with dispatch `calls=N` across both sinks across the entire run — no dropped frames, no thread starvation.

Tasks closed

- #130 [completed] Stage B.6 [DEFER] — migrate RX audio path to ported AAMix
- #132 [completed] Stage B.6.b — wire AAMix(1-input passthrough) into RX path
- Stages B.0–B.5 + B.6.a already closed in earlier sessions.

Methodology lessons recorded (for future Stage X work)

- 2-stage instrumentation (chain + dispatch) localized the defect to a single line of code on the SECOND bench. No speculation cycle, no convergence theatre.
- Operator-empirical rule held: after first B.6.b failure I was tempted to revert the AAMix port itself; the chain counters proved AAMix was correct in isolation, redirecting the dig to the wire-up surface where the bug actually was.
- Hard rule for next port: counters first when the symptom is "nothing happens". Instrumentation cost is trivial vs the alternative (multi-round revert/speculate cycle).
- "Do as Thetis does, Lyra-Native style" continues to hold — the first B.6.b attempt's `active=0x01` shortcut cost a revert cycle; the reference's `active=0 then SetAAudioMixState(activate)` pattern worked first time on the retry.

Memory + doc updates

- `MEMORY.md` index line for lyra-cpp-tx rewritten with "READ FIRST 2026-06-09 AM" header + branch/HEAD/backup/root-cause pointer.
- `project_lyra_cpp_tx.md` — new "■ READ FIRST 2026-06-09 AM resume pointer" block at top, plus full "State (2026-06-08 EOD)" section. Earlier state preserved below.
- `EXECUTION_PLAN.md` — new "PROGRESS TRACKING — STAGE B (aamix.c port) [SIDE-TRACK]" section with full B.0 → B.6.b-final checklist; new row in VERIFICATION LOG; footer "Status as of 2026-06-08 EOD" updated.
- This SESSION_LOG entry.
- PDF/DOCX regenerated via `tools/sync_execution_plan.py`.

Next session pointer

- **First action:** `git fetch origin` THEN `git status` + `git log --oneline -5 tx-rebuild origin/main` to surface divergence.
- **Next port target = OPERATOR'S CALL.** Stage B side-track done; candidates:
    - **Stage C:** `ilv.c` port (TX I/Q interleaver — pairs with the `SendpOutboundTx` stub already wired in `CMaster.cpp` Stage A)
    - **Stage D:** `xcmaster` pump body port
    - Or pivot back to Step 14 / Phase 3 wire-layer rebuild on `main` (previous-arc resume was TX-1 Component 7 — first end-to-end SSB voice TX)

Wire status

RX audio flows live through the ported AAMix dispatcher on the `tx-rebuild` branch's `HL2Stream::open()` path. TX path unchanged from earlier `tx-rebuild` state.

2026-06-06 (Saturday LATE-AFTERNOON — §1-C FULLY COMPLETE + RE-AUDIT CLEAN)

Branch: `tx-rebuild` HEAD = `169f1c2`. Build clean, `lyra.exe` relinked at 11:10:09. **16 commits shipped today** + TWO full multi-agent audit rounds. §1-C arc COMPLETE through Stage 4F.2 (FrameComposer dissolution). Final re-audit returned 37/37

PASS across 3 sections, zero remaining §6-Q3/Q5-class candidates, zero forbidden tokens in src/wire/.

What's reference-faithful now

• TX wire layer: 100% reference-faithful (all `_radionet`

fields in RadioNet; all reference file-scope globals as TU-scope statics in correct .cpp; Router + OutboundRing + FrameComposer all dissolved into free functions matching reference; symmetric socket-binding via metis_wire_bind() in both Ep6RecvThread + Ep2SendThread).

• §1.1 verdict: ■ → ■ PARITY (Stage 4E doc consolidation).

What's NOT yet reference-faithful (tracked, NOT in §1-C)

• **Task #73** — pre-existing forbidden-token violations

(Thetis / Console.cs / OpenHPSDR / PowerSDR) in 5 non-wire files: tci_server.{h,cpp}, mainwindow.cpp, hl2_stream.{h,cpp}, wdsp_native.h. Doc-style cleanup, ~1 hour, no impact on shipped wire-layer paths.

• **Wire-INERT** — nothing in HL2Stream::open() calls the

new wire-layer free functions yet (step 14 wire-up).

• **Task #114 deferrals** — EER mode, non-HL2

WriteMainLoop_generic, PeakFwdPower/PeakRevPower helpers. Hardware-blocker (no ANAN hardware to bench-test).

Step 14 scope (NOT trivial)

HL2Stream is 4203 lines total with its own monolithic `rxWorkerLoop` + `txWorkerLoop` member methods at `hl2_stream.cpp:921` + `:2437` (the OLD wire path). Step 14 must REPLACE these (per Rule 7 "Delete TX, don't refactor") with calls into the new wire-layer free functions:

1. `prn` global pointer assigned at session-open

2. `metis_wire_bind(socket, dest, dest_len)`

3. `outbound_init()` to size prn->outLRbufp/outIQbufp

4. `ForceCandC::prime(3, tx_freq, rx_freq)` synchronously

per §7 temporal-separation contract

5. `Ep6RecvThread::start(socket_fd)` → spins RX thread

reading from prn-> + WSAEventSelect

6. `Ep2SendThread::start(socket, dest, dest_len)` → spins

TX thread; no more composer / ring params (Stage 4F.2)

7. Migrate HL2Stream's existing telemetry/mic/IQ callbacks

to the new sink-based registration (Router::register_sink for IQ; Ep6RecvThread::set_*_sink for telemetry/mic/I2C)

Plan-first commit (R6). Bench-critical (first time the new wire layer actually emits datagrams to the radio). Multi-hour, possibly multi-commit. Wrong wire-up = broken RX/TX on real HL2 hardware.

Resume after step 14

Operator-bench verification of RX-side audio + telemetry on real HL2+ hardware. Then TX-side bench (CW/SSB + Palstar power readings). Then Task #73 cleanup (small, doc-only, can happen anytime).

Wire status

Wire-INERT. All wire-layer components reference-faithful but not connected to runtime.

2026-06-06 (Saturday AFTERNOON — §1.1 REVERT COMPLETE — historical)

Branch: `tx-rebuild` HEAD = `79fe4ec`. Build clean, `lyra.exe` relinked at 10:45:10. **TWELVE commits shipped today** + 6-agent comprehensive TX audit done. ALL stages of the §1-C sweep complete in code (§1.1 networking- infrastructure exclusion fully reverted). Only Stage 4E (PARITY_CHECKPOINTS doc consolidation) remains before step 14 wire-up.

Today's commits (newest first)

- `79fe4ec` §1-C Stage 4D — OutboundRing dissolves to free functions (§1.1 revert COMPLETE)
- `1d73bea` §1-C Stage 4C — Ep2SendThread reads from prn->OutBufp + g_fpga_write_bufp
- `f1031da` §1-C Stage 4B.1 — correct FPGAReadBufp scope (file-scope, not _radionet)
- `26acea3` §1-C Stage 4B — Ep6RecvThread reads from prn-> + WSAEventSelect
- `e157301` §1-C Stage 4A — add networking-infrastructure fields to RadioNet
- `d88ddd6` §1-C Stage 3 — OutboundRing wait-all via condition_variable
- `2b78d9e` §1-C Stage 2A — Router class → struct + free functions
- `0e2a375` §1-C Stage 1 — small reverts (wb_enable, push timeout, diag counters)
- `50b713b` §6-B nit — drop metis_write_frame null-guard
- `3bfd248` §6-B + §7 — TU-scope MetisFrame + ForceCandC populate

§1.1 revert summary

All `_radionet` fields the original §1.1 exclusion (signed ■ 2026-06-04) had punted to wire-layer class members now live in RadioNet as members exactly per reference:

- Buffer pointers: `RxBuff`, `TxReadBufp`, `outLRbufp`, `outIQbufp`, `OutBufp` (+ `ReadBufp` declared P2-future-use)
- RX seq counters: `cc_seq_no`, `cc_seq_err` (P2-only; declared in RadioNet but HL2-inert)
- Thread handles: `hReadThreadMain`, `hWriteThreadMain`, `hKeepAliveThread` (std::thread types)
- Init semaphores: `hReadThreadInitSem`, `hWriteThreadInitSem` (std::counting_semaphore<1>)
- Outbound sync (cv-collapsed mirror of 4 reference HANDLE pairs): `cv_outbound` + `mu_outbound` + 4 bool flags + `outbound_stop`
- WSA event + waitable timer (Win32-only): `hDataEvent`, `wsaProcessEvents`, `hTimer`, `liDueTime`
- Networking config ports: `p2_custom_port_base`, `base_outbound_port`

Reference-FILE-SCOPE globals (NOT in `_radionet`) live as TU-scope statics in their respective wire-layer .cpp files per the §6-B precedent:

- `wire/MetisFrame.cpp`: `g_metis_out_seq_num` (was Ep2SendThread::out_seq_num_)
- `wire/Ep6RecvThread.cpp`: `g_metis_last_recv_seq`, `g_seq_error`, `g_seq_seen`, `g_control_bytes_in[5]`, `g_fpga_read_bufp`
- `wire/Ep2SendThread.cpp`: `g_fpga_write_bufp`

Wire-layer classes dissolved or reduced:

- `Router`: class → plain struct + free functions (Stage 2A)
- `OutboundRing`: class → namespace-scope free functions operating on `prn->...` (Stage 4D)
- `Ep6RecvThread`: still a class (thread orchestrator), but

all data moved to `prn->...` or TU-scope statics

• `Ep2SendThread`: same — thread orchestrator, data moved out

Remaining — Stage 4E (doc-only)

Consolidate the §1-C entry in `docs/architecture/PARITY_CHECKPOINTS.md`:

• Single §1-C entry covering Stages 1+2A+3+4A-D + the

earlier §6-B + §7

• §1.1 verdict amended from ■ OPERATOR-APPROVED DEVIATION

to ■ PARITY (now genuinely byte/structure-faithful)

• Sign-off block + commit trailer

Resume after Stage 4E

Phase 2 step 14 — wire `outbound_init()` + `metis_wire_bind()` + `ForceCandC::prime` + `Ep6RecvThread::start` + `Ep2SendThread::start` into `HL2Stream::session_open()` in the correct temporal sequence.

Wire status

Wire-INERT today. All wire-layer components built but nothing in `HL2Stream::session_open` calls them yet.

2026-06-06 (Saturday MID-DAY — STAGE 3 CHECKPOINT — historical)

Branch: `tx-rebuild` HEAD = `d88ddd6`. Build clean, `lyra.exe` relinked at 09:54:09. Five commits shipped today + 6-agent comprehensive TX audit done. Stages 1+2A+3 of the §1-C sweep DONE; Stages 2B+4 remain (bundled — §5.10 WSAEventSelect lives in `_radionet` fields per reference, so it lands with §1.1 revert).

Today's commits (newest first)

• `d88ddd6` §1-C Stage 3 — OutboundRing wait-all via condition_variable

• `2b78d9e` §1-C Stage 2A — Router class → struct + free functions

• `0e2a375` §1-C Stage 1 — small reverts (wb_enable, push timeout, diag counters)

• `50b713b` §6-B nit — drop metis_write_frame null-guard

• `3bfd248` §6-B + §7 — TU-scope MetisFrame + ForceCandC populate

What got fixed to match reference (Stages 1+2A+3)

• §1.3: `std::atomic wb_enable` → `volatile long` (matches `network.h:160`)

• §6.10: OutboundRing producer push — drop bounded 5s try_acquire_for + push_timeouts_*; unbounded `acquire()` mirroring reference Inbound/obbuffs

• §6.12: Ep2SendThread `datagrams_sent_/send_errors_` diagnostic counters dropped (no reference counterpart)

• §5.9: `class Router` → plain struct + free functions (`xrouter/register_sink/set_control_word/set_call_count/router_instance`) + file-scope `g_routers[]` array, mirroring reference `router.c` structure verbatim

• §6.9/6.11/6.14: OutboundRing wait-all via `std::condition_variable` + `bool` flags + single mutex (was 4 binary_semaphores + polling) — mirrors reference `WaitForMultipleObjects(2, hsendEventHandles, TRUE, INFINITE)` atomic wait-all, no polling, fully interruptible

Remaining — Stage 4 (THE BIG ONE)

§1.1 networking-infrastructure revert + §5.10 WSAEventSelect bundled.

§1.1 (signed ■ 2026-06-04) excludes from RadioNet a list of fields that the reference puts INSIDE `_radionet`. Under strict "do as reference, period" §1.1 itself is a §6-Q3/Q5- class candidate. Stage 4 puts these fields back into RadioNet:

• Buffer pointers: `RxBuff`, `TxReadBufp`, `ReadBufp`, `OutBufp`, `outLRbufp`, `outIQbufp`

• Thread/sem/event handles: `hReadThreadMain`, `hsendEventHandles[2]`, `hobbuffsRun[2]`, etc.

• Waitable timer + WSA event: `liDueTime`, `hDataEvent`, `wsaProcessEvents`

• RX seq counters: `cc_seq_no`, `cc_seq_err`

• Networking ports: `p2_custom_port_base`, `base_outbound_port`

Then Ep6RecvThread + Ep2SendThread + OutboundRing read from `prn->...` instead of their own members. §5.10 (recv() → WSAEventSelect) folds in because `hDataEvent` + `wsaProcessEvents` are `_radionet` fields.

Scope: multi-hour, multi-commit. Planned sub-stages:

• 4A: Add fields to RadioNet (additive, no behavior change)

• 4B: Ep6RecvThread refactor + WSAEventSelect mechanism

• 4C: Ep2SendThread refactor (reads buffers from RadioNet)

• 4D: OutboundRing — buffers + sync primitives move into

RadioNet; class may dissolve into free functions

• 4E: PARITY_CHECKPOINTS §1-C entry finalized + §1.1 amendment

Resume after Stage 4 completes

Phase 2 step 14 — wire `ForceCandC::prime` + `metis_wire_bind` + WSAEventSelect setup into `HL2Stream::session_open()` in the correct temporal sequence.

Wire status

Wire-INERT today. All wire-layer components built but nothing in `HL2Stream::session_open` calls them yet.

2026-06-06 (Saturday MID-DAY — AUDIT PAUSE BOOKMARK — historical)

Branch: `tx-rebuild` HEAD = `50b713b`. Build clean, `lyra.exe` relinked at 09:27:22. All §6-B + §7 work shipped + audited 20/20 PASS this morning.

Done this morning (2 commits, ALL like reference)

• `3bfd248` §6-B parity correction sweep + §7 ForceCandC populate

    • Hoisted `metis_write_frame` to `wire/MetisFrame.{h,cpp}` TU-scope

    • TU-scope `g_metis_out_seq_num` (was `Ep2SendThread::out_seq_num_`)

    • TU-scope socket/dest globals (were Ep2SendThread members)

    • DROPPED `Ep2SendThread::send_lock_` mutex (no reference counterpart)

    • §1.1 row split: RX seq stays / TX seq moves

    • `ForceCandC::prime` + `prime_pass` byte-for-byte from `networkproto1.c:106-139`

• `50b713b` §6-B nit — dropped `metis_write_frame` null-guard (no reference

counterpart; audit-caught NOTE-D)

• Two parallel audits run: parity (20/10 PASS) + rule-compliance (10/10 PASS)

Pause reason

Operator-requested **comprehensive TX audit** of ALL work since the rebuild started (30 commits since `0348238` Phase 0 mapping doc 2026-06-04). Sections to audit against the reference:

• §1 RadioNet (`2171fb9`/`dcbf9d1`)

• §2 RbpFilter / RbpFilter2 (`228ffd8`)

• §3 HPSDR family + dispatch globals (`f3ccc51`/`55ae658`/`782e65d`)

• §4a/§4b-1/§4b-2/§4c FrameComposer 19-case dispatch

(`93e1511`/`5fc4192`/`4446916`/`f06a796`)

- §5 / §5-A Ep6RecvThread + Router (`c8baa63`/`a6be425`/`b988177`/`6bbf449`)
- §6 / §6-A Ep2SendThread + OutboundRing (`4c580a1`/`89ab298`)
- §6-B + §7 already audited this morning (skipping re-audit)

Resume point after audit

Resume here: **Phase 2 step 14 — wire-up** `ForceCandC::prime` **+** `metis_wire_bind` **into** `HL2Stream::session_open()` in the correct temporal sequence (bind → prime → Ep6 start → Ep2 start).

Wire status

Wire-INERT today. Nothing in `HL2Stream::session_open` calls the new wire-layer components yet. Step 14 (next commit after audit closes) flips the wire on.

2026-06-05 (Friday EOD)

Project: lyra-cpp TX Wire-Layer Rebuild (Task #112) — branch `tx-rebuild`

Done today

- **§5 EP6 Receive Path** — `Ep6RecvThread` + `Router` (`xrouter/twist`) populated (`c8baa63`)
- **§5-A 9-fix parity correction** (`a6be425`) including ■ CRITICAL RX-audio bug (IQ unpack scale 48 dB too quiet — caught before any consumer wired)
- **§5-A doc completeness** (`b988177`)
- **§6 EP2 Send Path** — `Ep2SendThread` + `OutboundRing` populated (`4c580a1`)
- **§6-A 6-fix parity correction** (`89ab298`) — MMCSS priority, per-family dispatch, CW bit-shift revert to verbatim, CW state per-sample reads, `binary_semaphore` UB guard, Rule 24 inline token
- 4 audit passes total (2 per §5 / §6) — all PASS after corrections

Where we are

Branch `tx-rebuild` HEAD = `89ab298`. §5 + §6 of the wire-layer rebuild are **shipped + doubly-verified** (parity-audit + rule-audit + correction sweeps + re-audits, all clean). Wire-layer is still wire-inert — no Phase 1 code path instantiates the new components.

Next up

§7 — `ForceCandC` **priming + ANY remaining wire-layer pieces.** Reference: `networkproto1.c::ForceCandCFrame` (lines 134-139) + `ForceCandCFrames` (lines 106-132). Small commit — just the 3-frame priming burst that runs at session start before the EP2 send loop begins.

After §7 completes the §1c-listed wire-layer skeletons, the next phase is **Phase 2 step 14 — wire-up**: instantiate Ep6RecvThread + Ep2SendThread + OutboundRing + Router + ForceCandC + FrameComposer into HL2Stream's session-open path.

Pending / parked

- **Task #114** TX-policy plumbing — ATT-on-TX, panadapter offset, PA-enable safety, plus the §6 FIXMEs: `WriteMainLoop_generic` for non-HL2, EER mode (§6.4), `PeakFwdPower`/`PeakRevPower` helpers (§5 telemetry cases 0x08/0x10)
- **No hardware bench yet** — gateware-watchdog kill-test, real-antenna TX, foot-switch HW-PTT all parked until Phase 2 wire-up + verification
- **Don't push to GitHub** (standing rule)

Commit summary (today)

```
89ab298 fix(wire): §6-A parity correction sweep — 6 fixes
4c580a1 feat(wire): populate §6 Ep2SendThread + OutboundRing
b988177 docs(parity): §5-A Lyra-native additions table — 2 audit-flagged rows
a6be425 fix(wire): §5-A parity correction sweep — 9 fixes
c8baa63 feat(wire): populate §5 Ep6RecvThread + Router
```

Standing rules in effect

- Rule 24 "always verify against reference" — applied per commit, audit-verified
- 2026-06-05 directive "reference = make Lyra the same" — driving all Q-decisions
- §6 EER mode + non-HL2 dispatch + Peak power helpers all deferred to Task #114 with FIXMEs
- Operator hardware = HL2+/AK4951; no ANAN P1/P2 hardware available

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Session log started 2026-06-05. Older entries (if any) below this line as they accumulate.